



Vehicle-Track Measurement Technologies

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Engineering Systems, Inc.

B O S T O N

Summary



Overview of Measurement Systems

Detailed Evaluation of Select Technologies

Simulations Using Measurement Data

Overview of Measurement Systems

There are six basic categories of measurement system.



Categories of Measurement Systems



1) Mounted on Vehicle to measure the Vehicle.



Image from Amsted Digital Solutions



Image from Hitachi Rail (Perpetuum)



Categories of Measurement Systems

2) Mounted on Track to measure the Track.

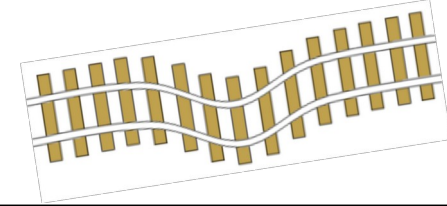


Image from FTS Tools



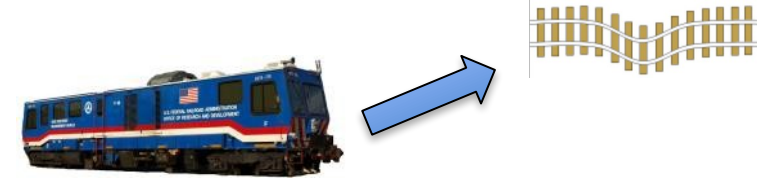
Image from Voestalpine



Categories of Measurement Systems



3) Mounted on Vehicle to measure the Track.



Manned



Autonomous



B O S T O

Light Vehicles



Image from Holland



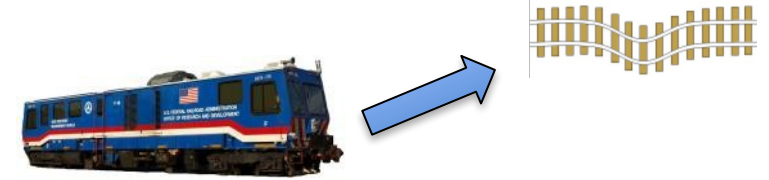
Image from RailPod



Categories of Measurement Systems



3) Mounted on Vehicle to measure the Track.



Pushcart



Image from Amberg

Drones

Unmanned Aerial Vehicle (UAV)
Unmanned Aerial System (UAS)



Multicopter
Multirotor



Vertical Take-Off
and Landing (VTOL)



Categories of Measurement Systems

4) Mounted on Track to measure the Vehicle.

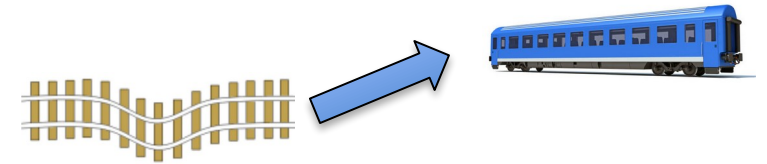


Image from L.B. Foster



Image from Hexagon, Business Unit CALIPRI



Categories of Measurement Systems



5) Mounted on Vehicle to measure the Vehicle & Track.

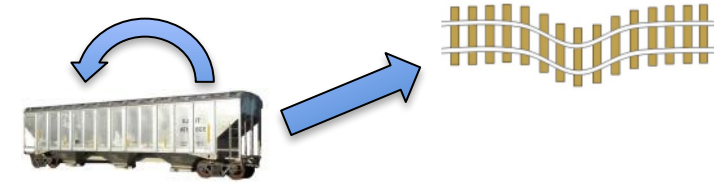


Image from ENSCO Rail, Inc.

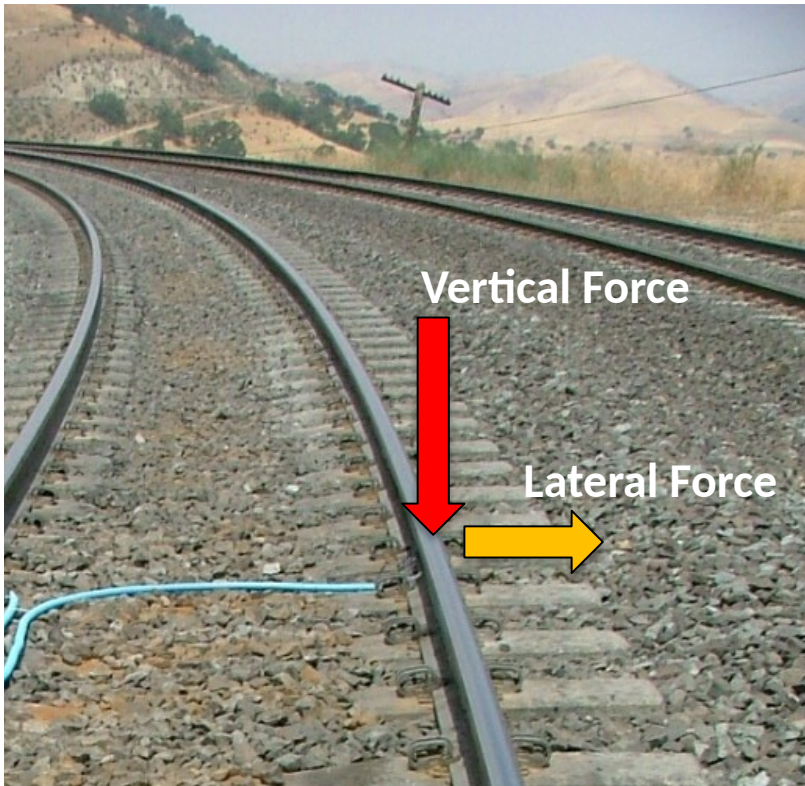
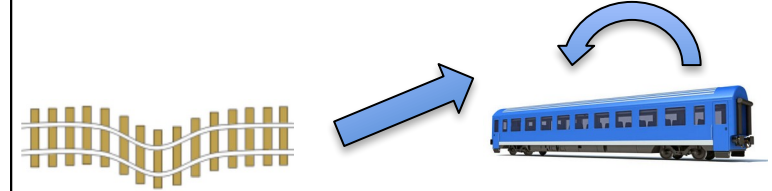


Image from ENSCO Rail, Inc.

Categories of Measurement Systems



6) Mounted on Track to measure the Vehicle & Track.



Truck Performance Detectors

Also known as “L/V Detector”

Common Uses:

Vehicle condition monitoring
Superelevation assessments
New Vehicle Fleet Assessments



Track Measurement

Track Geometry Measurement

Rail Profile Measurement System

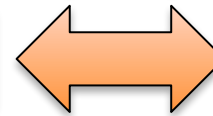
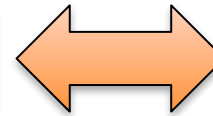
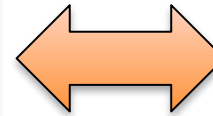
V/TI Monitor Axle Impact

Vehicle Measurement

**Truck Condition Monitor
(TBOGI)**

Wheel Profile Detector

Wheel Impact Load Detector



Geometry Measurement

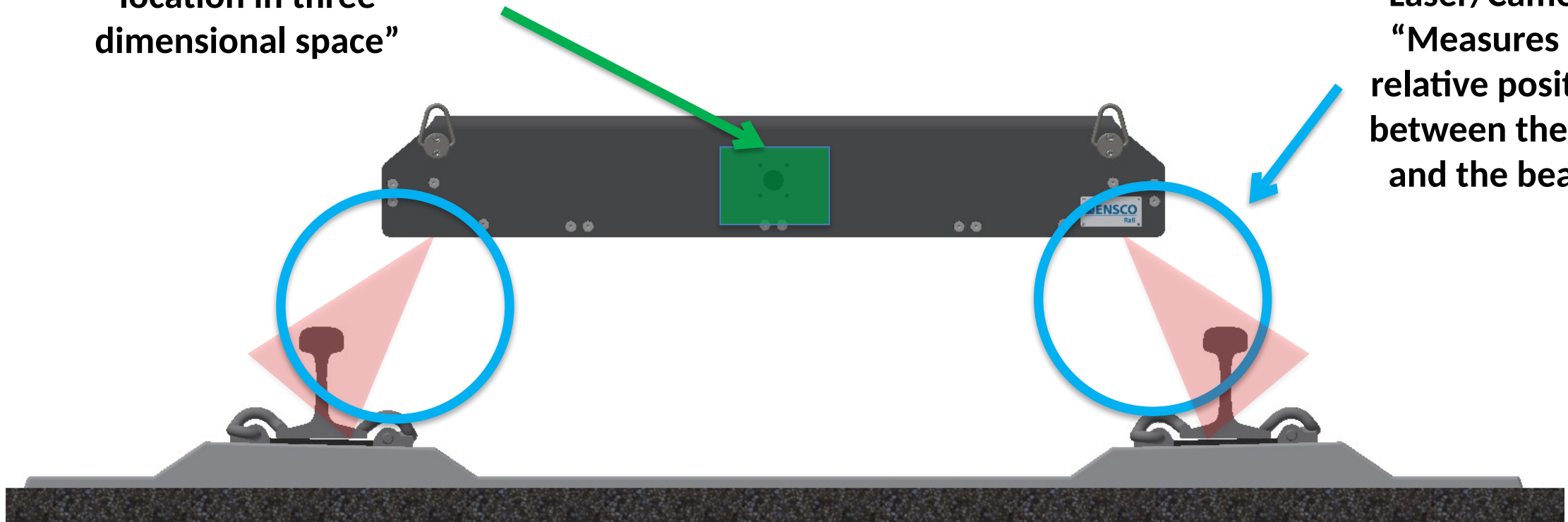


Track Geometry Measurement System



Inertial Package
“Measures the beam
location in three
dimensional space”

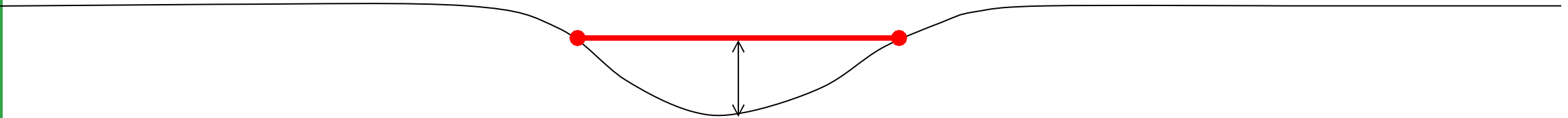
Laser/Cameras
“Measures the
relative positions
between the rails
and the beam”



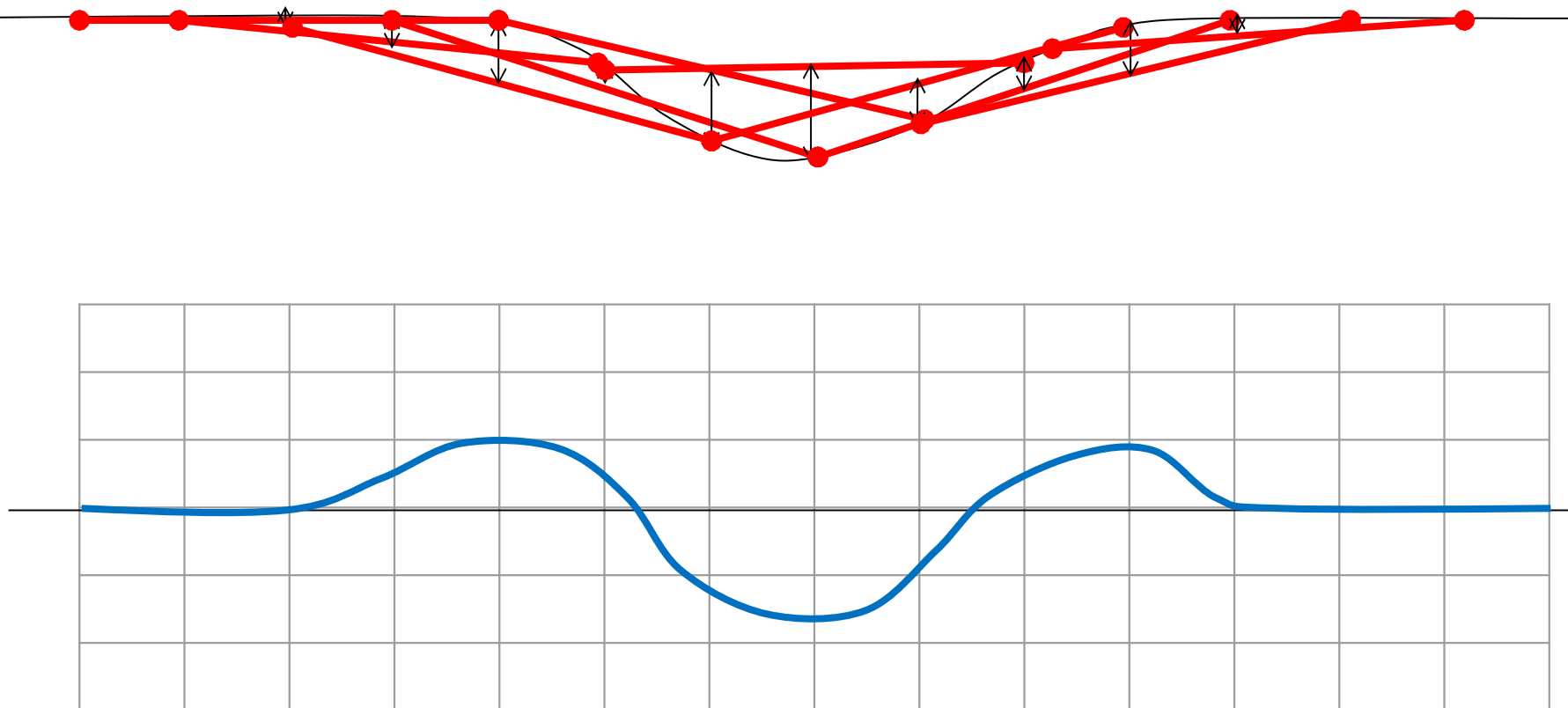
Space Curve vs. MCO

Space Curve is intended to represent the true geometry of Profile or Alignment conditions.

Mid-Chord Offset is used to quantify Profile or Alignment conditions.



Space Curve vs. MCO

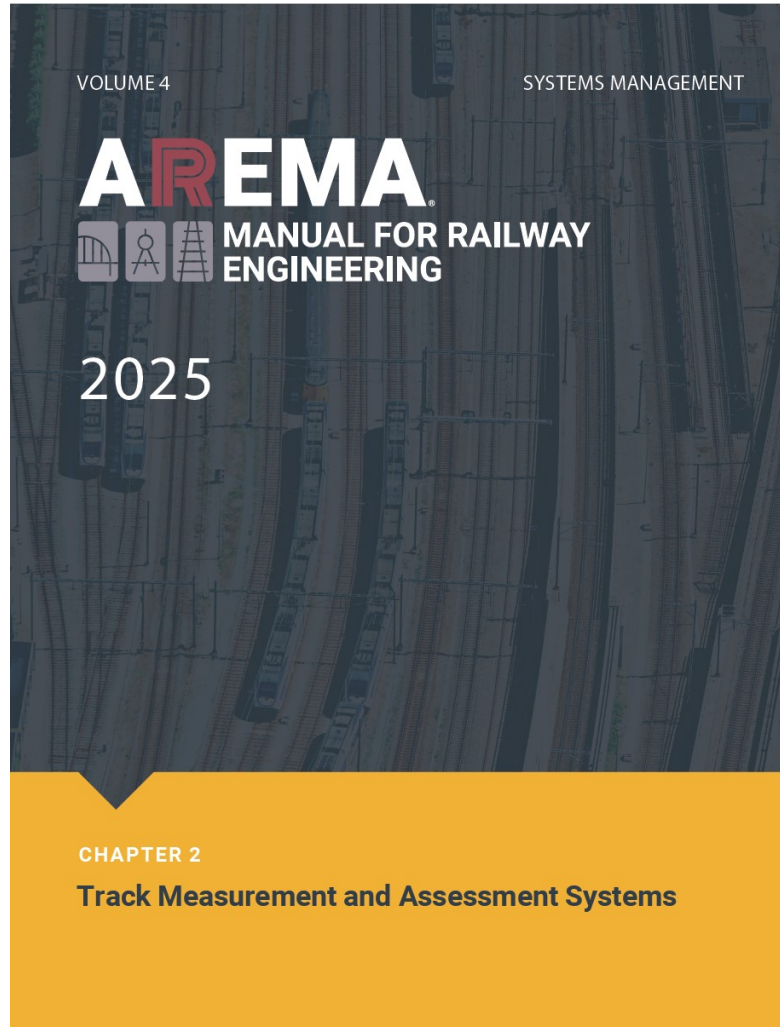


Common Profile and Alignment Chord Lengths:

10 ft
31 ft
62 ft
124 ft



Space Curve vs. MCO



AREMA Chapter 2 now has a great definition of Space Curve and MCO (Part 3 Track Geometry Measurement)

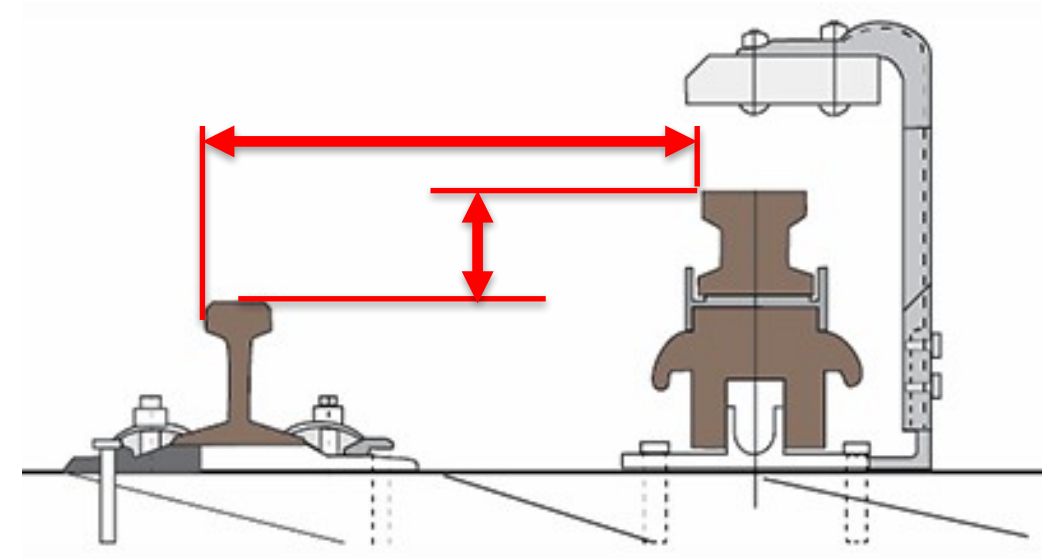
Also New:
SECTION 9.3 RECOMMENDED PRACTICE
FOR USING TRACK GEOMETRY
MEASUREMENT SYSTEM DATA FOR
VEHICLE/TRACK INTERACTION
MODELING



Third Rail Geometry Measurement



Images from ENSCO Rail, Inc.



Catenary Geometry Measurement

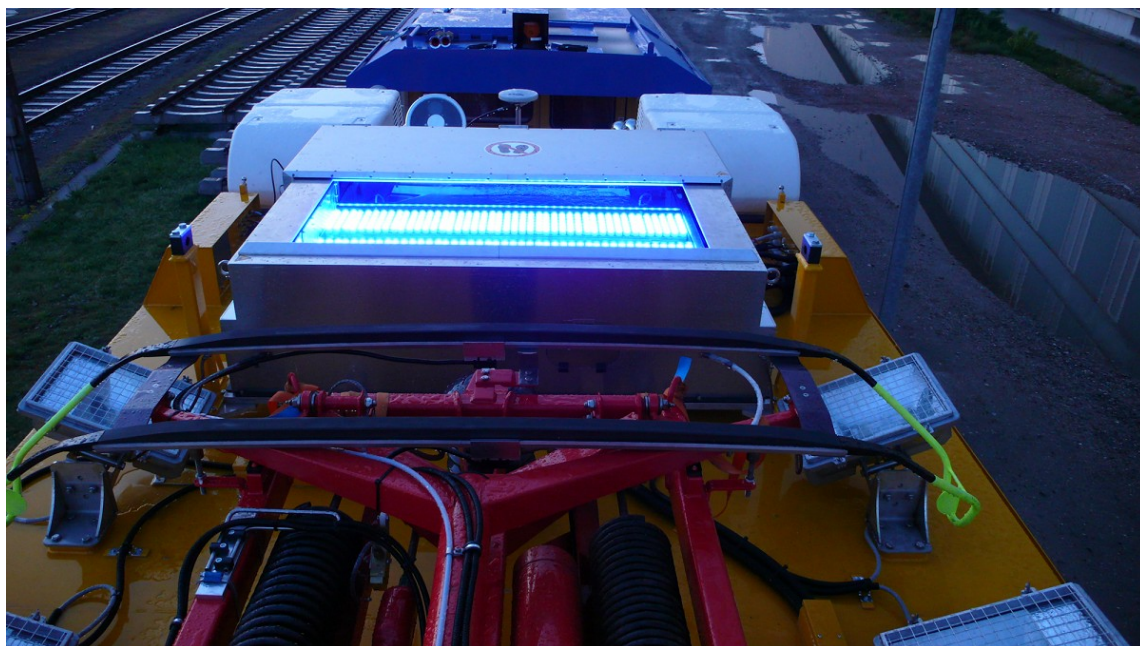
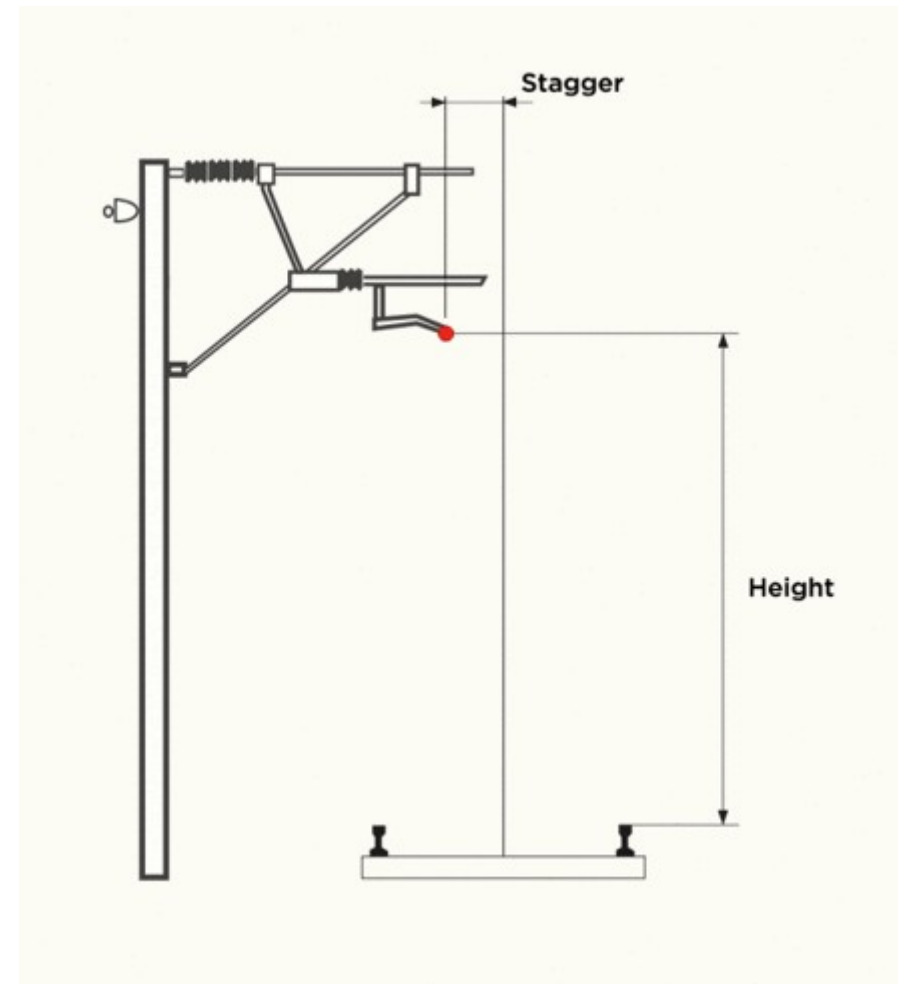


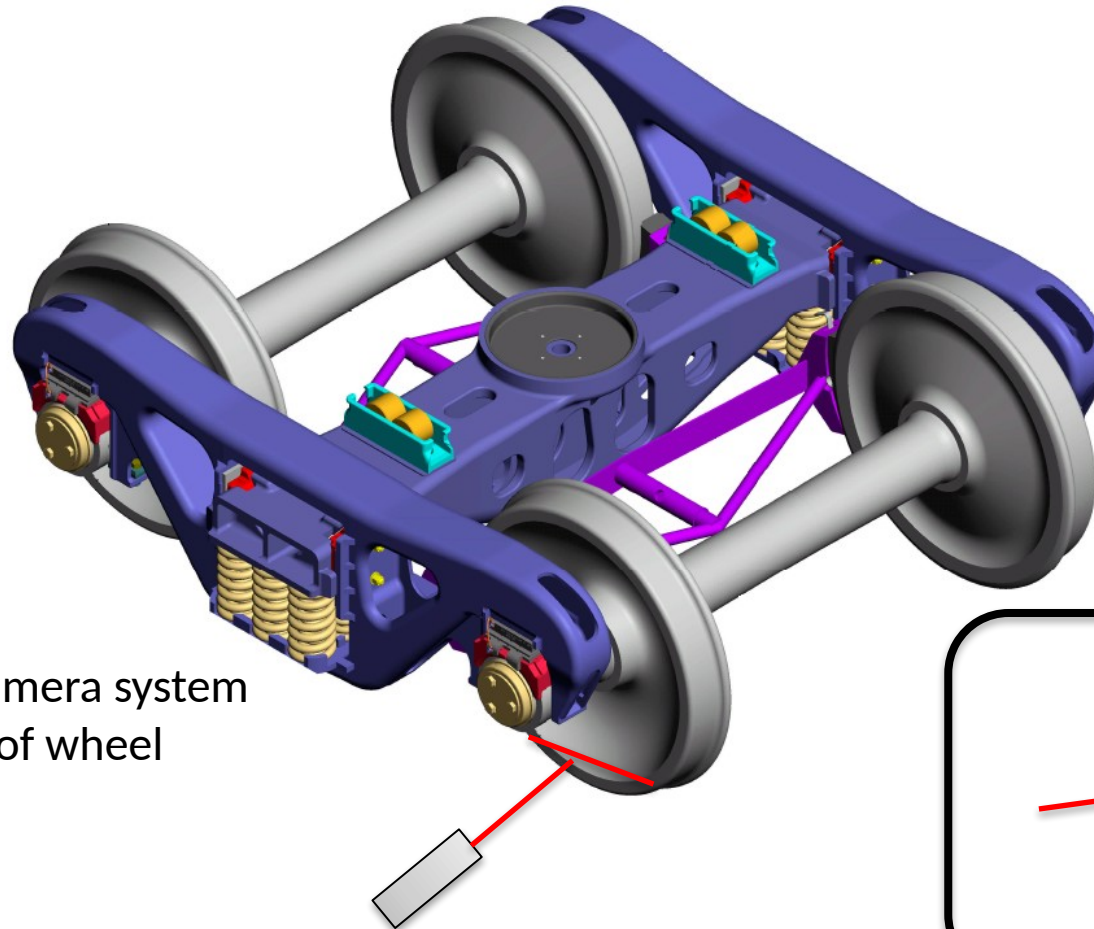
Image from Fraunhofer



Ref: <https://tc.at.es/english-measuring-the-height-and-stagger-of-a-contact-wire/>

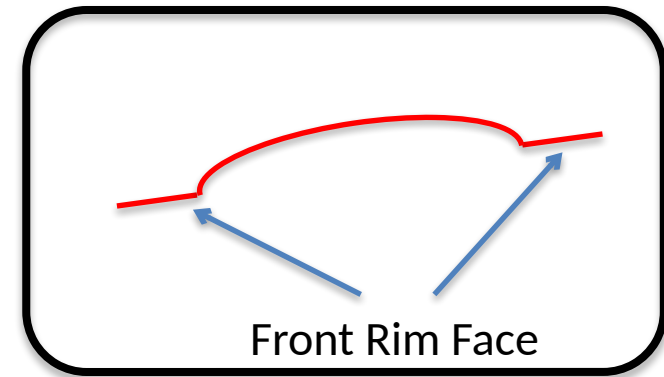


Truck Condition Monitor (T-BOGI)



Wayside Laser/Camera system measures profile of wheel plate.

Laser

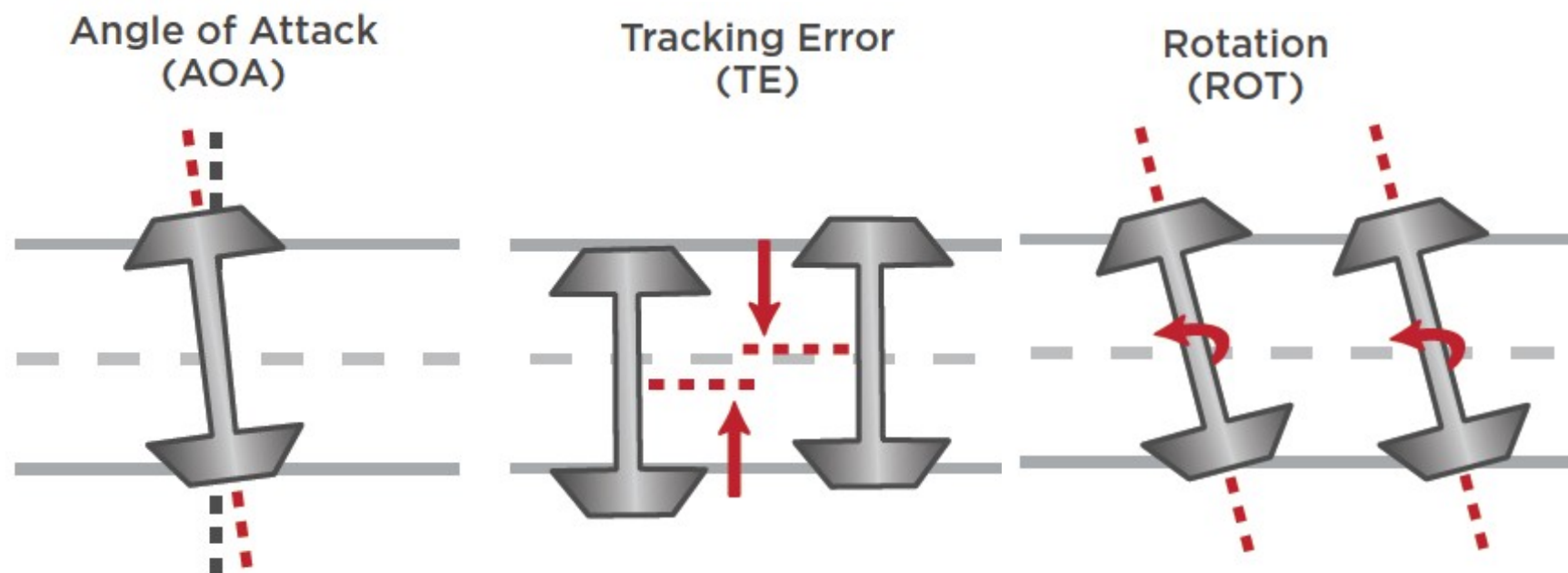


Front Rim Face

Image from Standard Car and Truck Maintenance Manual
<http://www.sctco.com/pdf/Section1.pdf>



How does a truck condition monitor work?

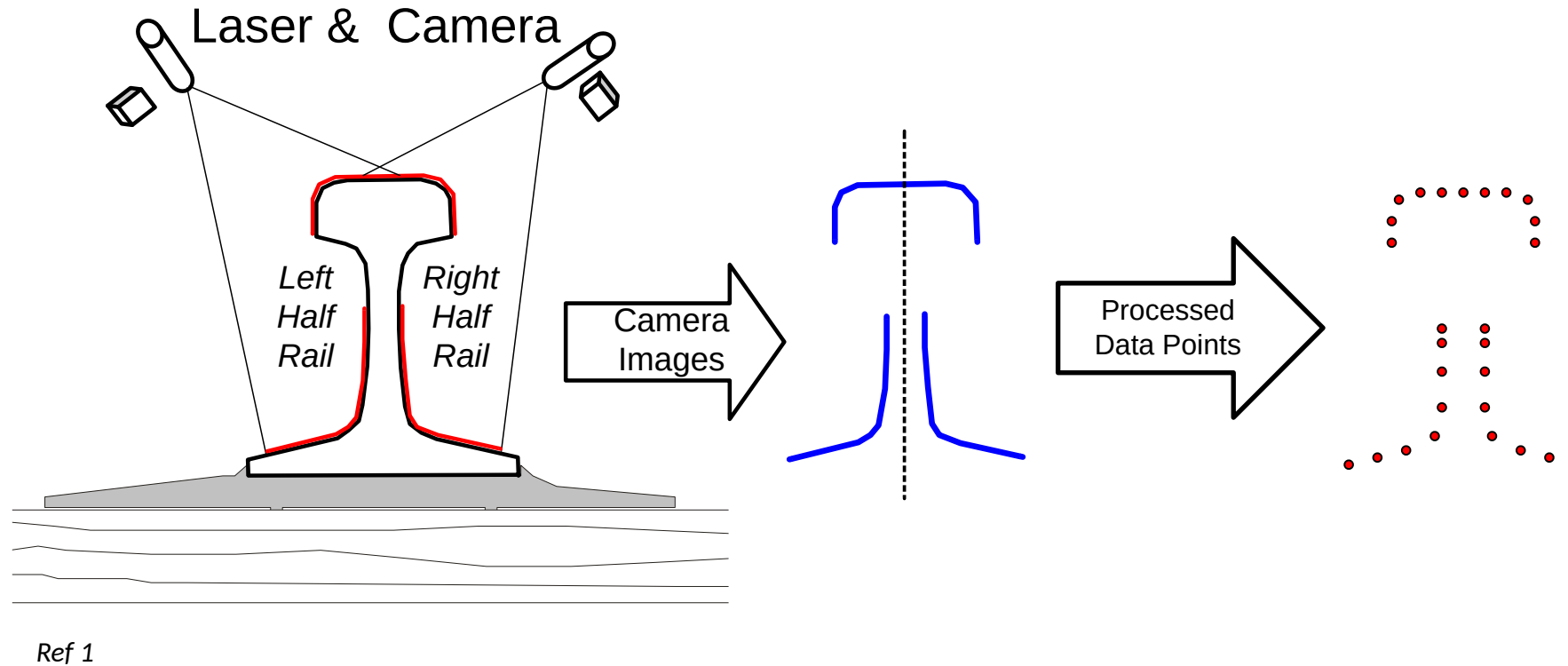


Images from Wayside Inspection Systems
http://wid.ca/sites/default/files/brochures/TBOGI/WID_TBOGI_Brochure_US.pdf

Profile Measurement



Laser Triangulation Measurement



Laser Triangulation Measurement

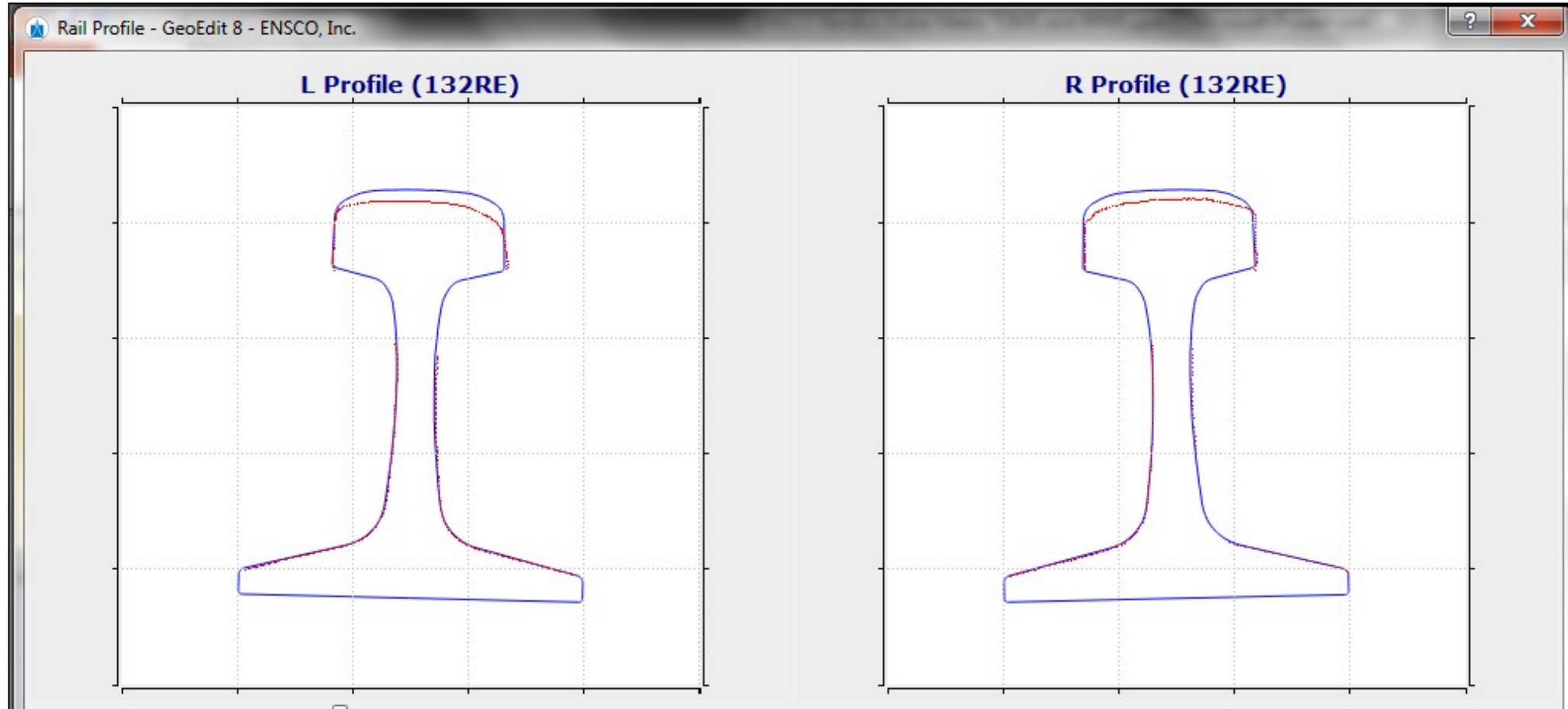
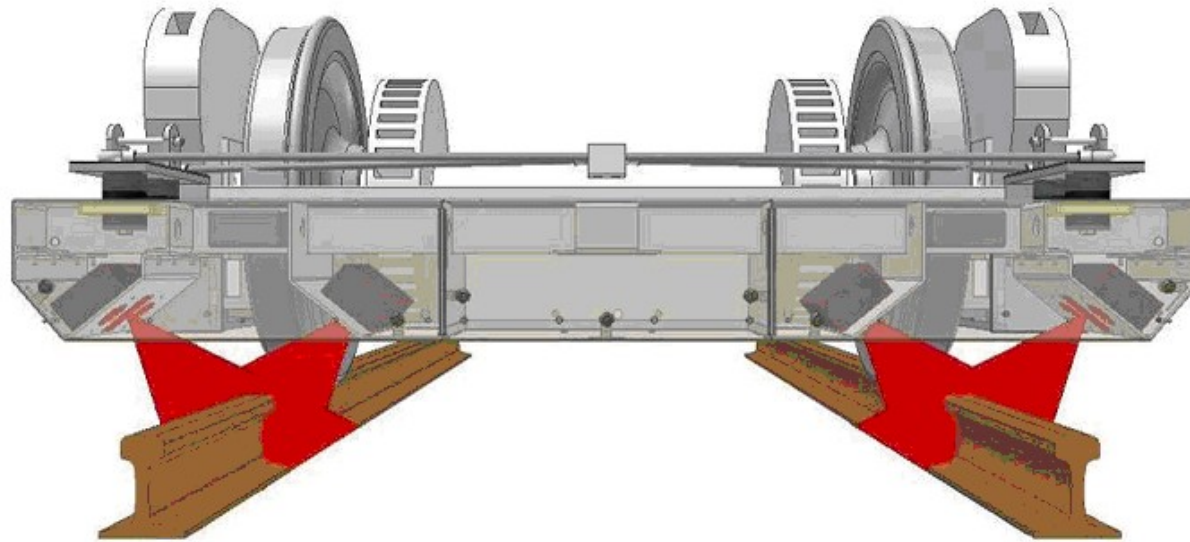


Image from ENSCO Rail, Inc.



Rail Profile Measurement System



Images from ENSCO Rail, Inc.

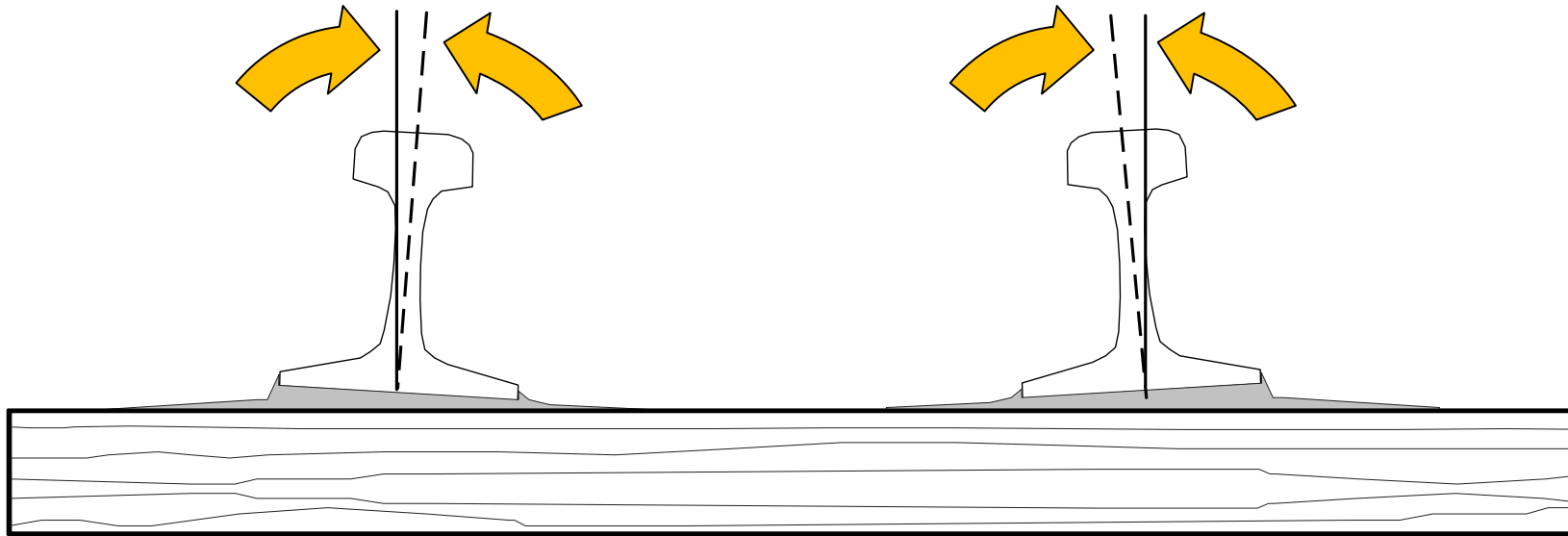


Absolute Measurements:

(Doesn't Require Template)



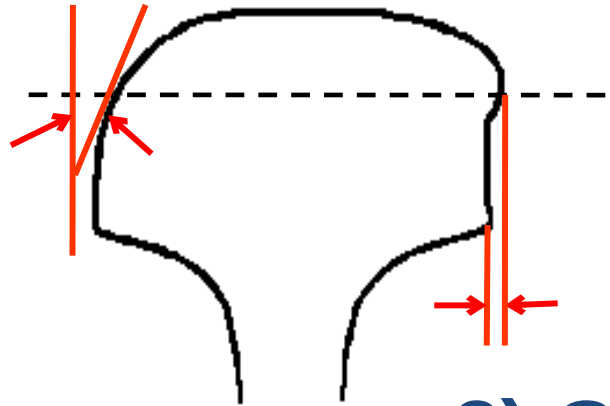
1) Cant (Deg)



Absolute Measurements:

(Doesn't Require Template)

**2) Gage
Face Angle
(Deg)**



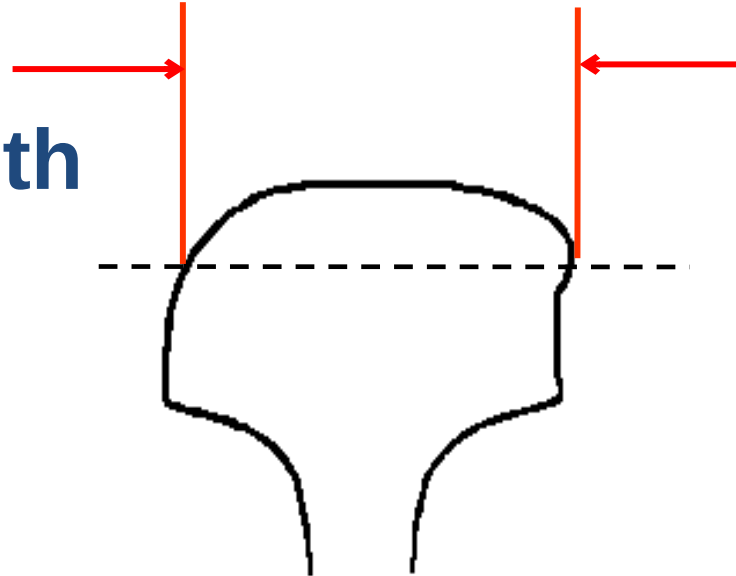
**3) Gage/Field
Side Lip (in, mm)**



Absolute Measurements:

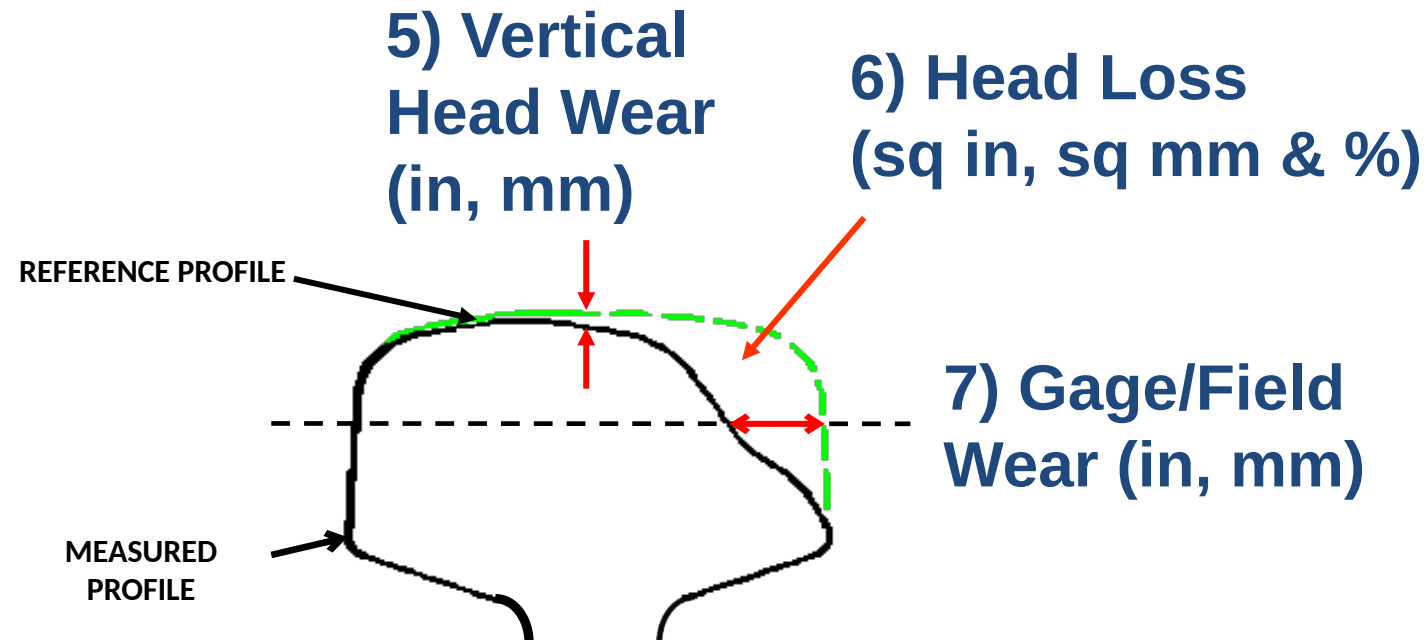
(Doesn't Require Template)

4) Head Width (in, mm)



Relative Measurements:

(Does Require Template)



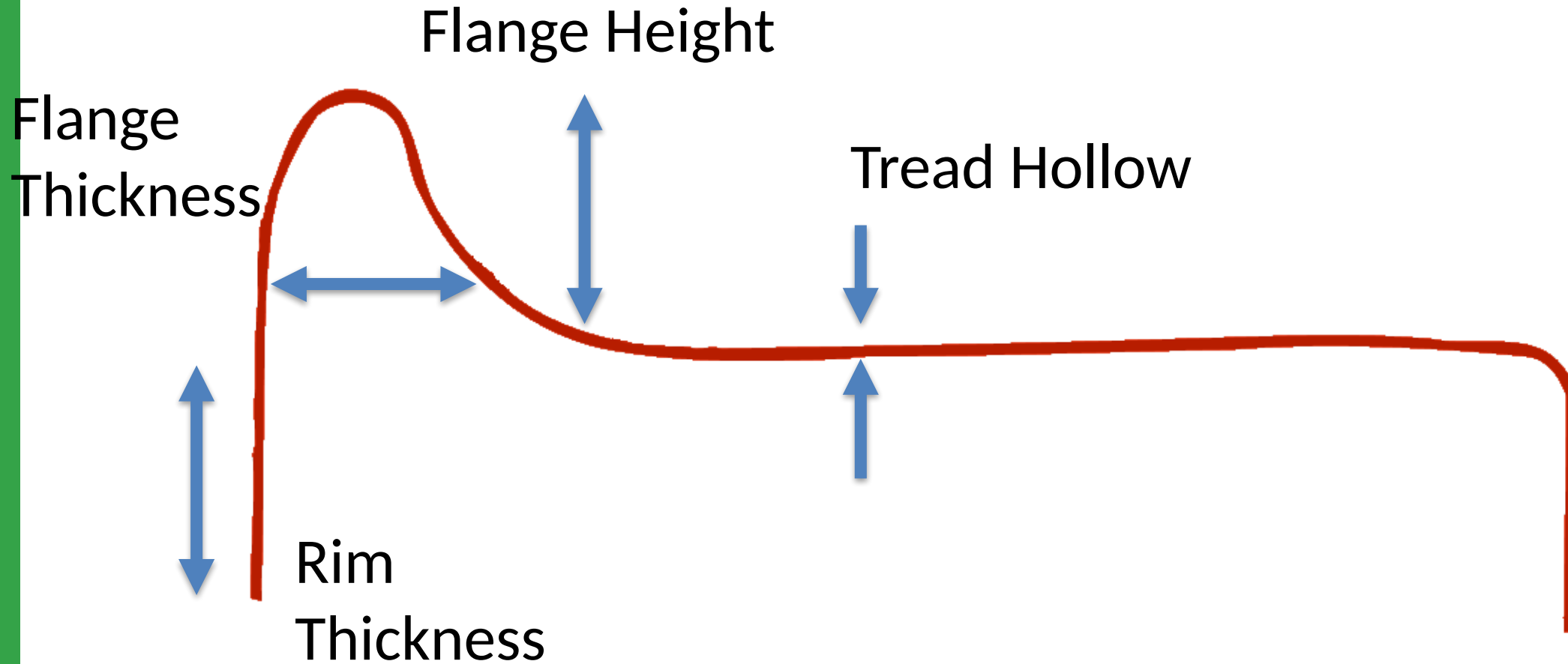
Wheel Profile Detector



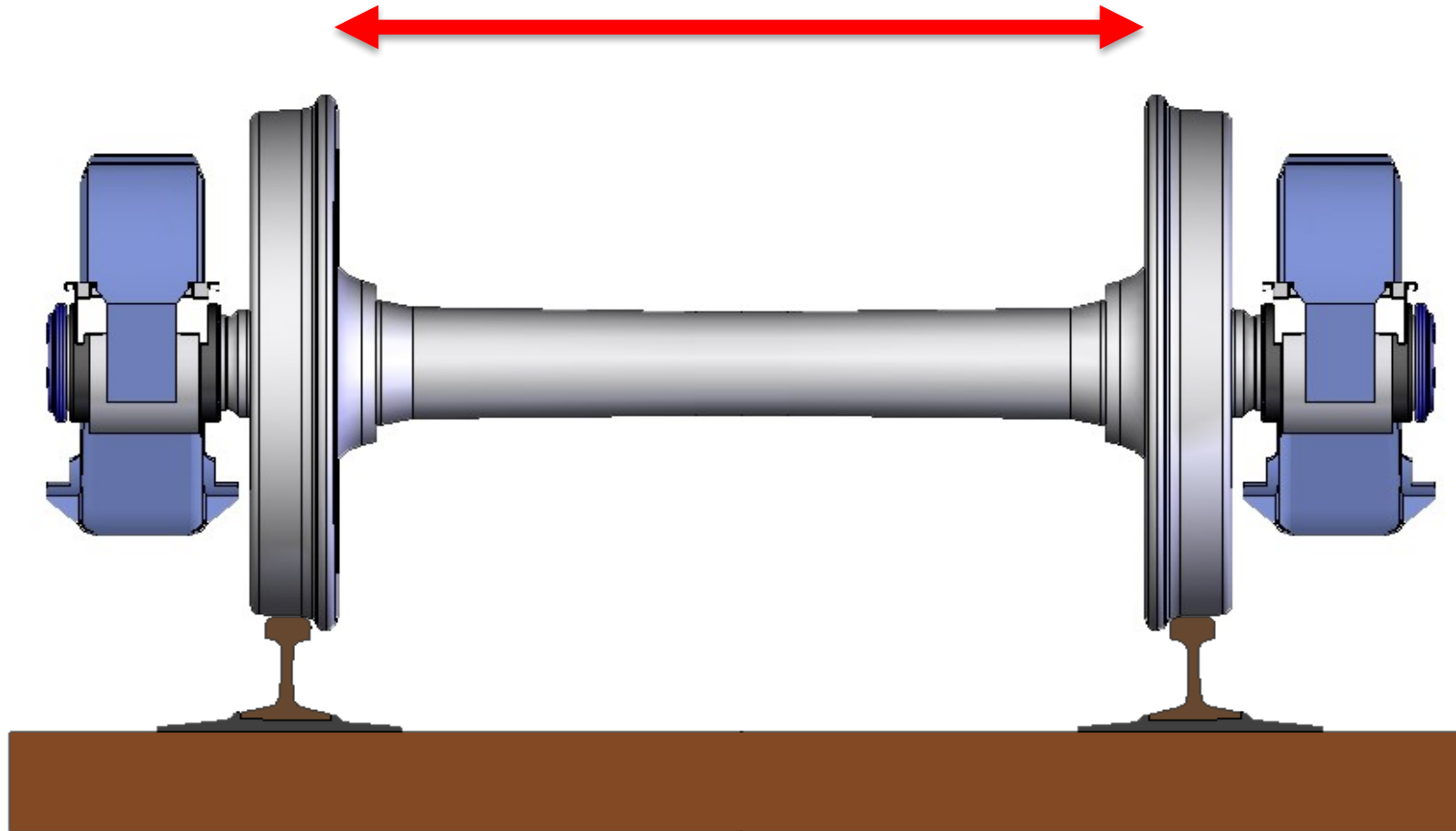
Image from KLD Labs, Inc.



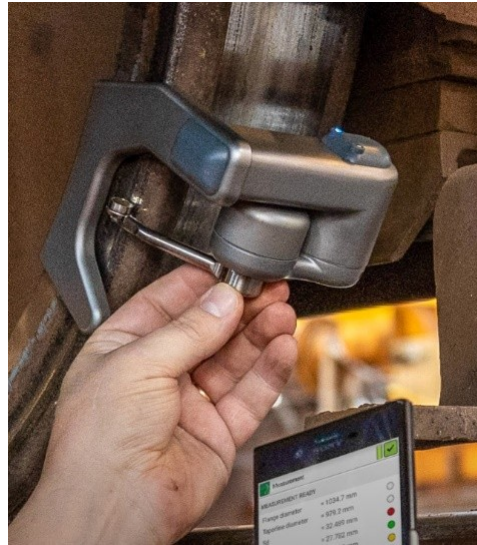
Image from Hexagon, Business Unit CALIPRI



Back-to-Back



Handheld Wheel and Rail Profile Measurement



Images from Greenwood Engineering

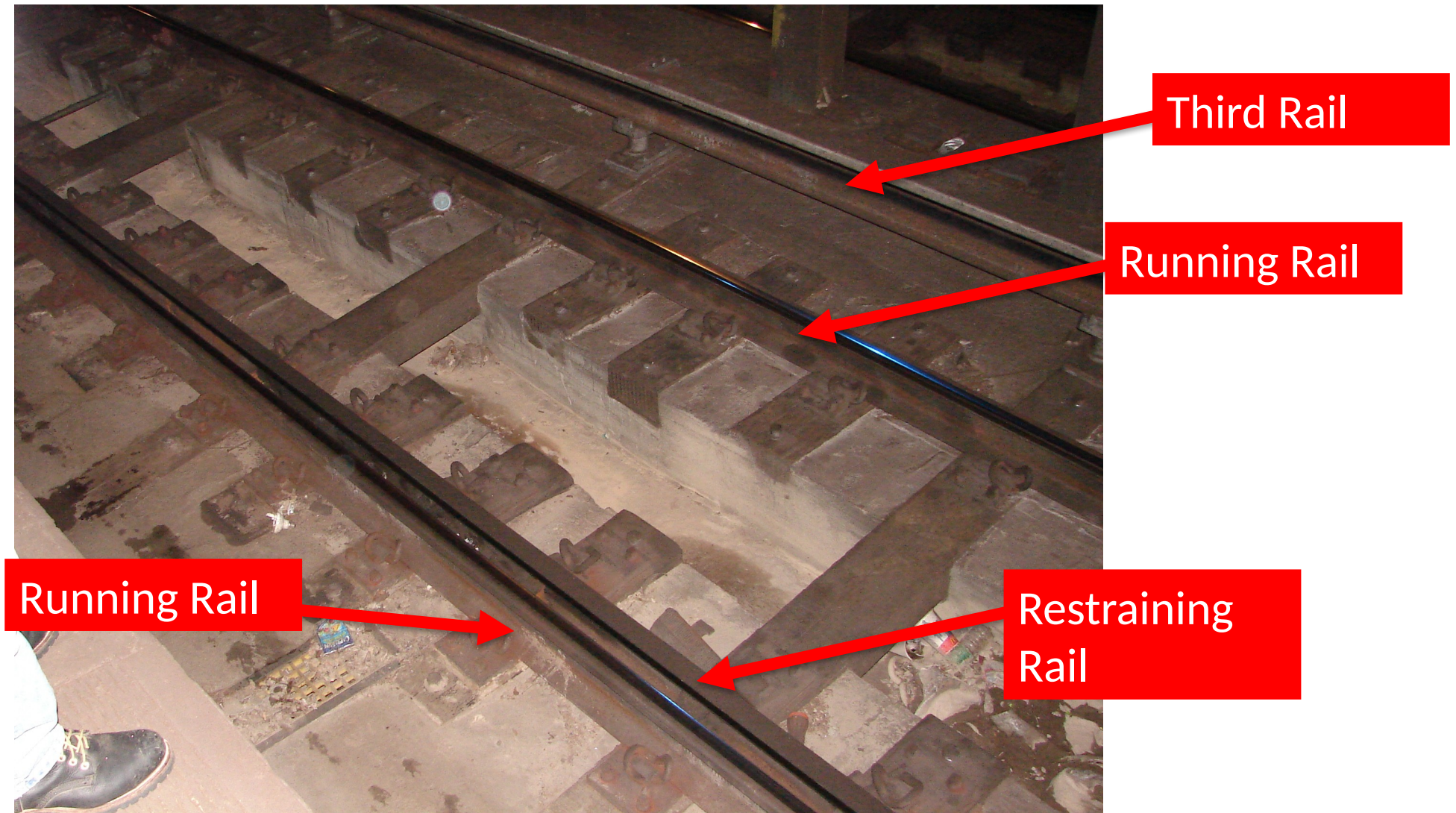


Images from Hexagon, Business Unit CALIPRI

Restraining Rail Case Study for Wheel and Rail Profiles

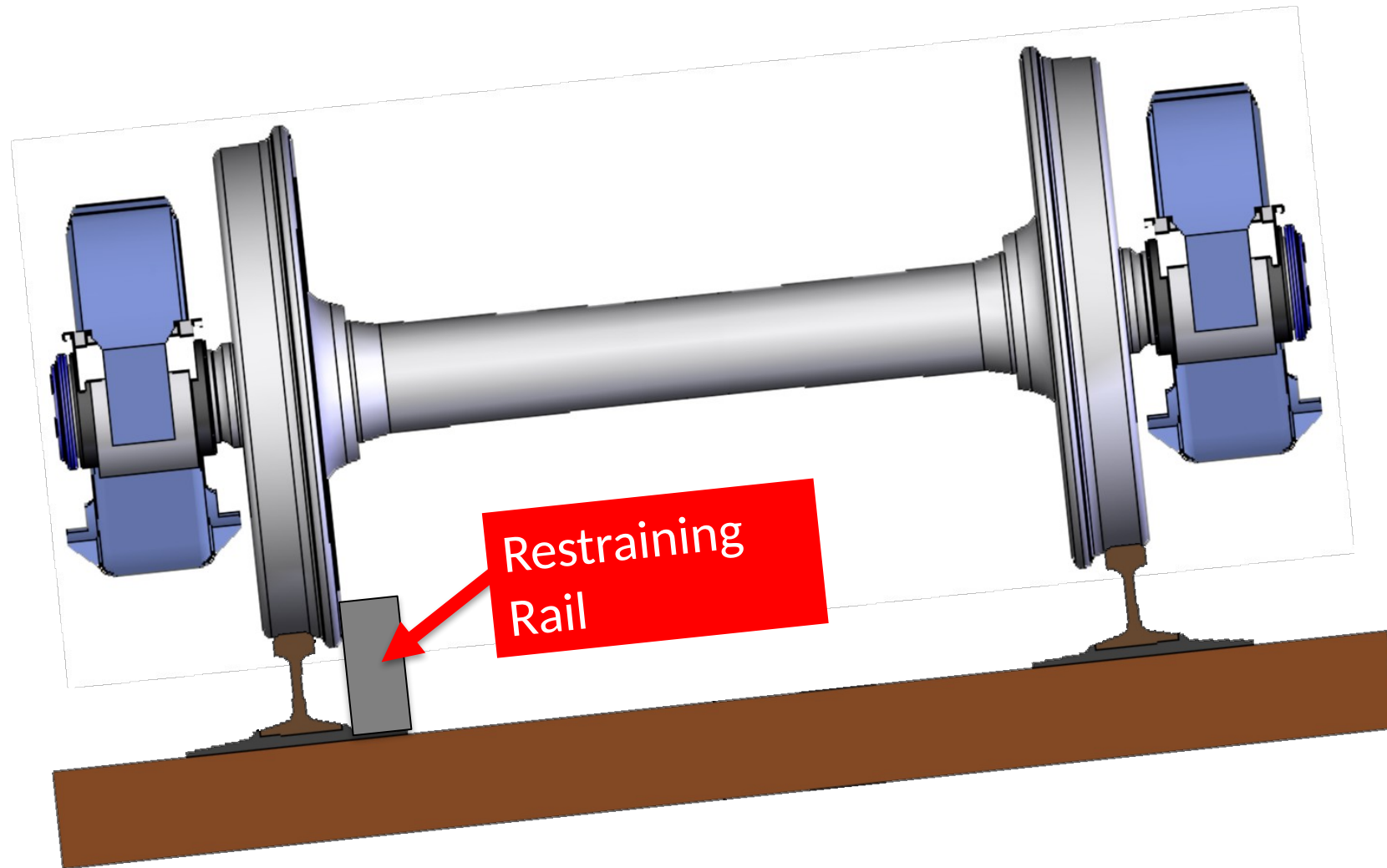


Restraining Rail Case Study



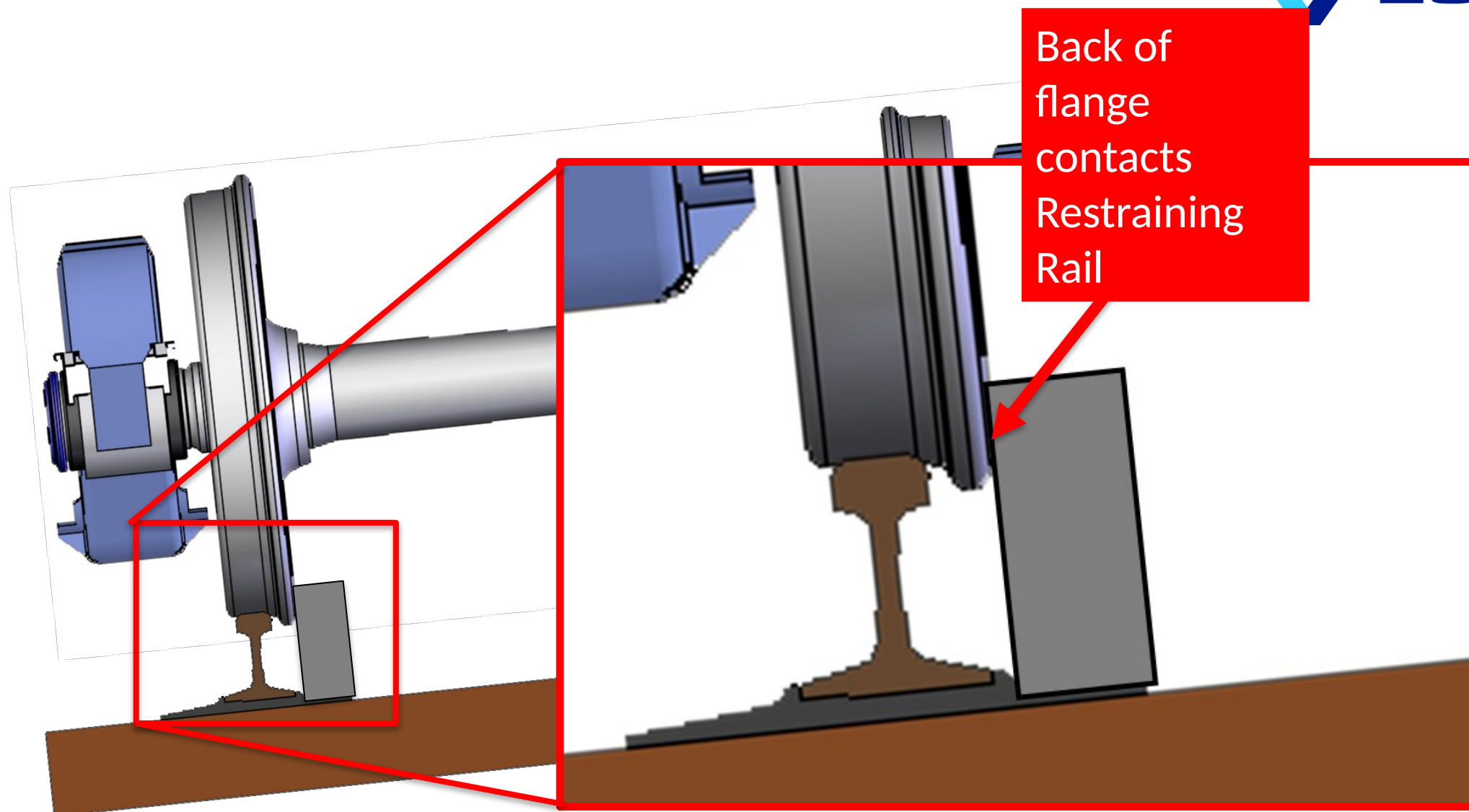
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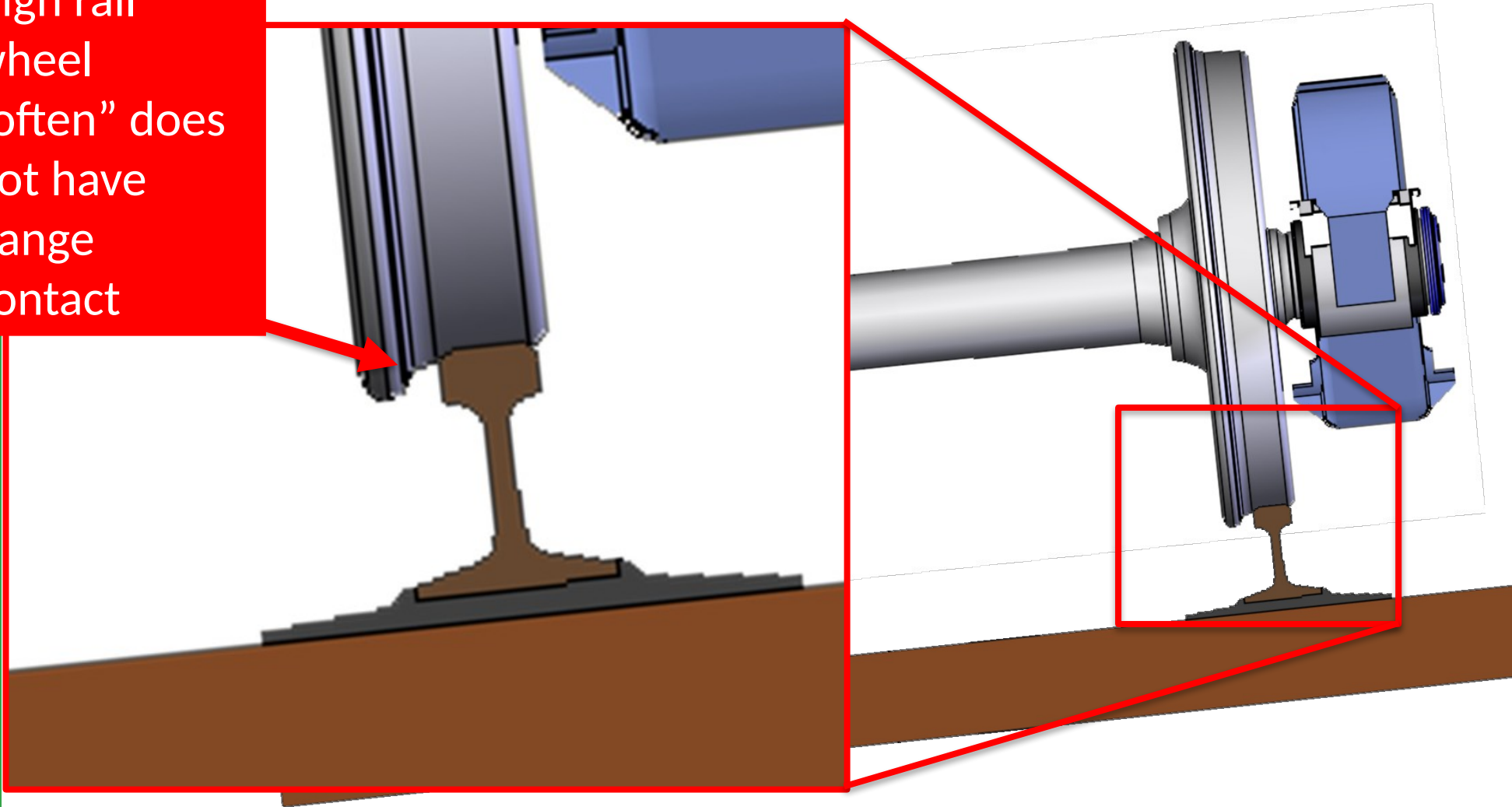


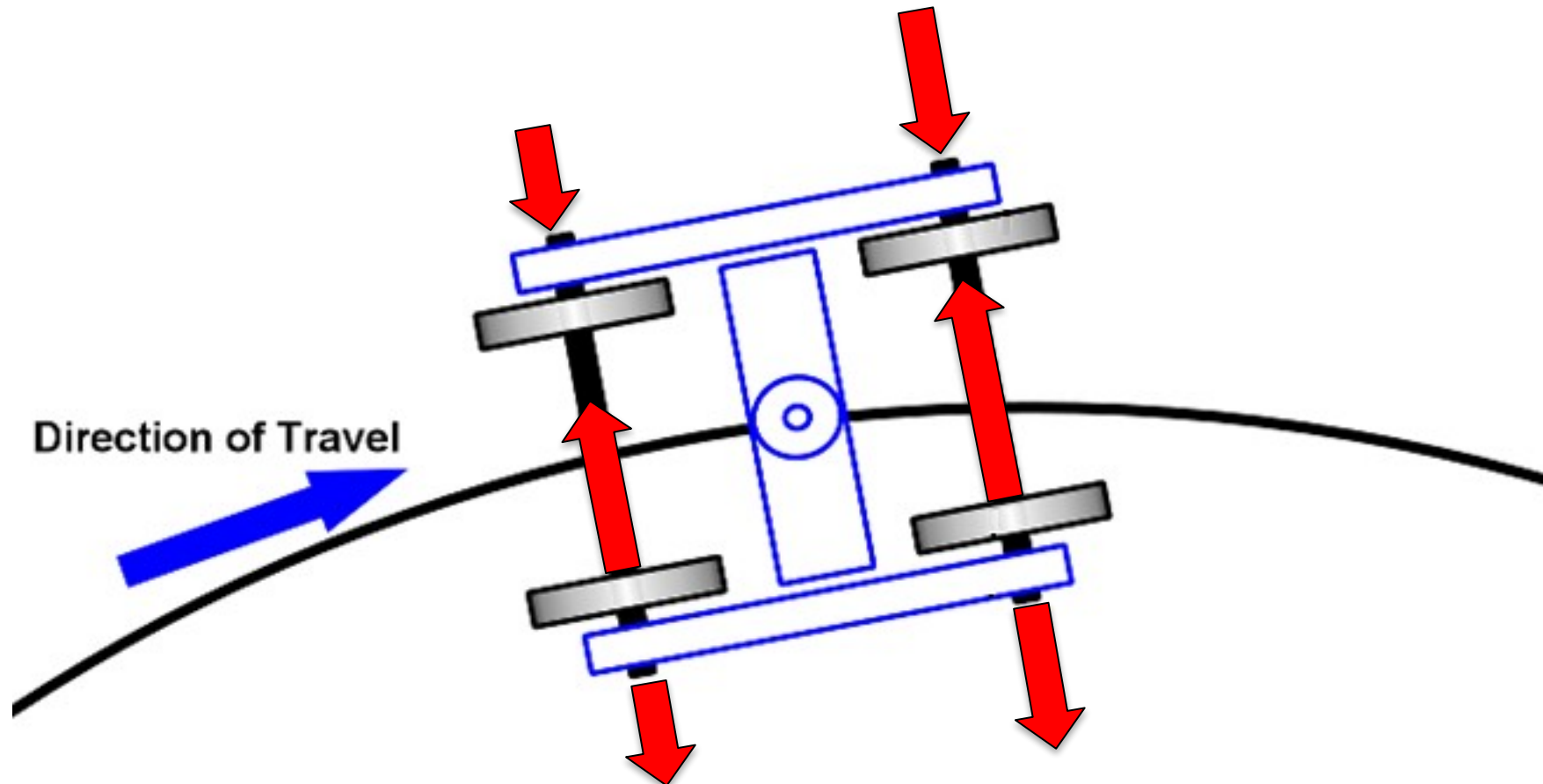
B O S T O





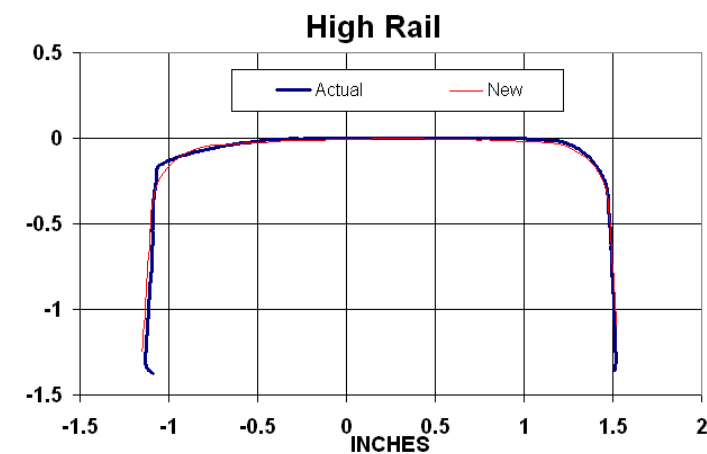
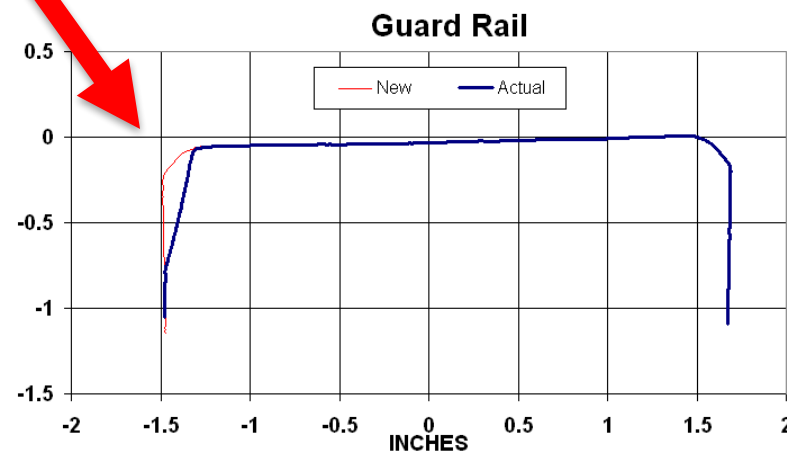
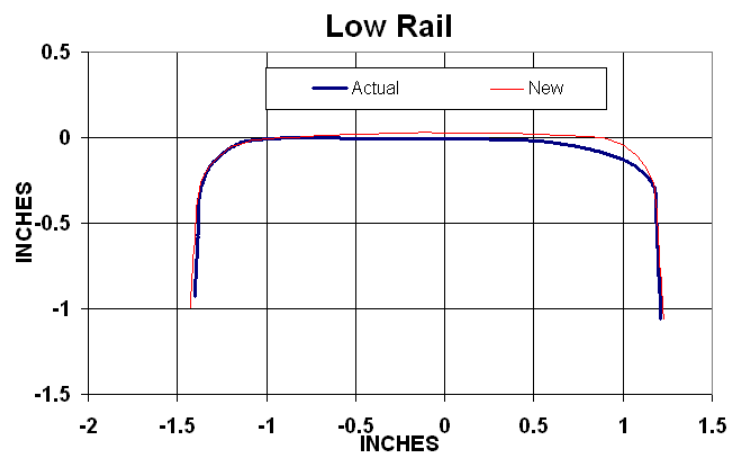
High rail
wheel
“often” does
not have
flange
contact





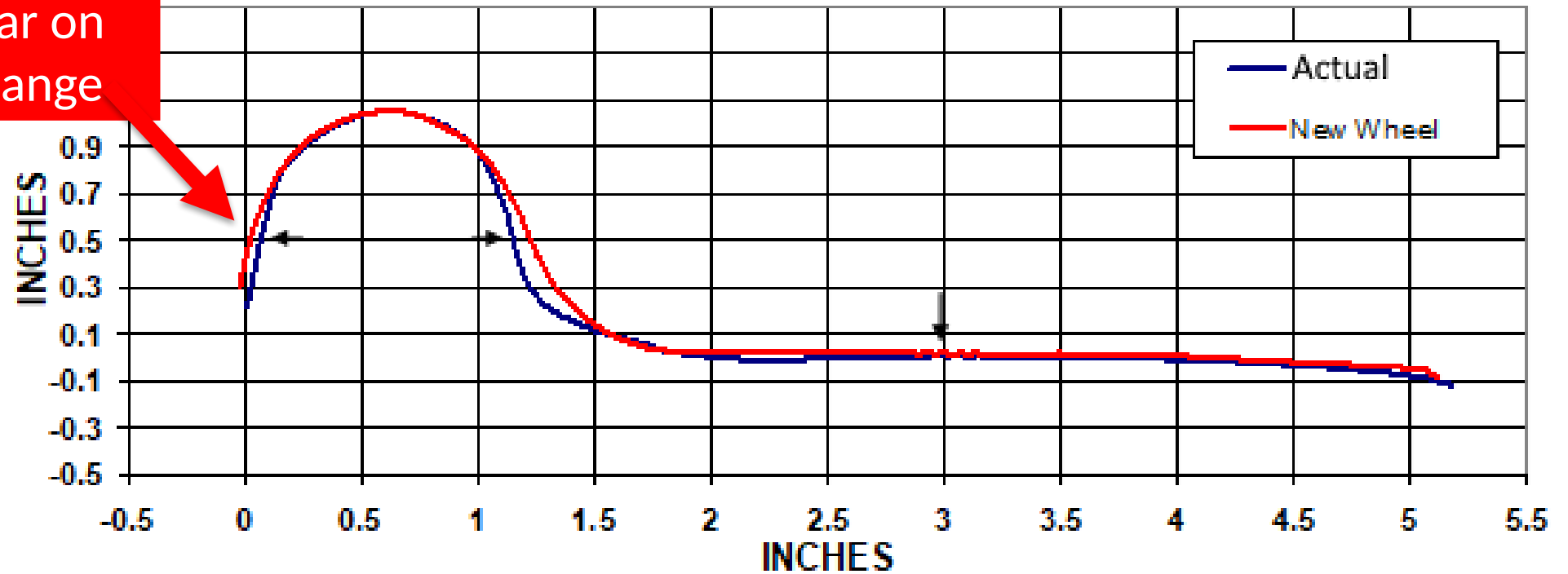


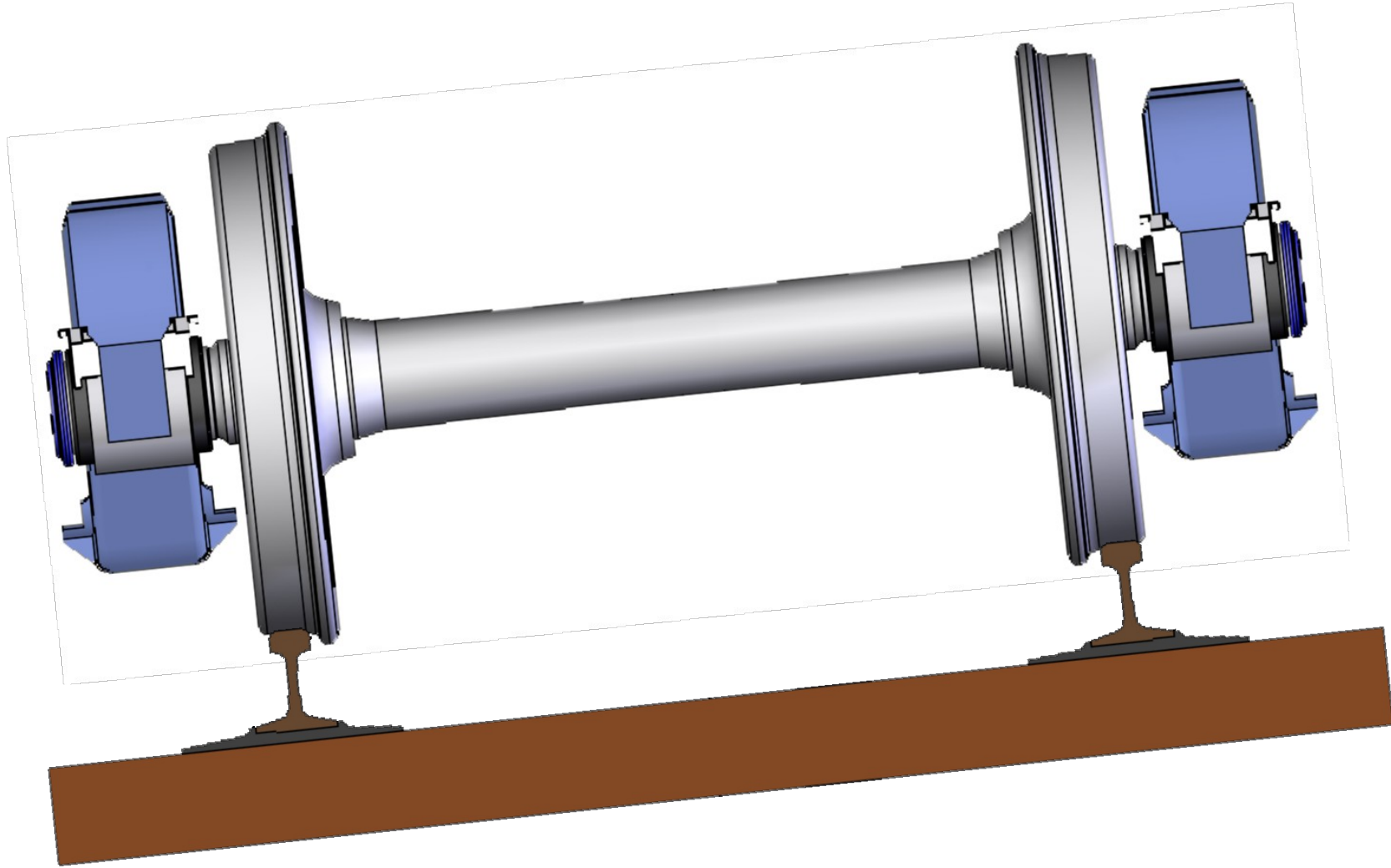
Sloped Wear on Guard Rail





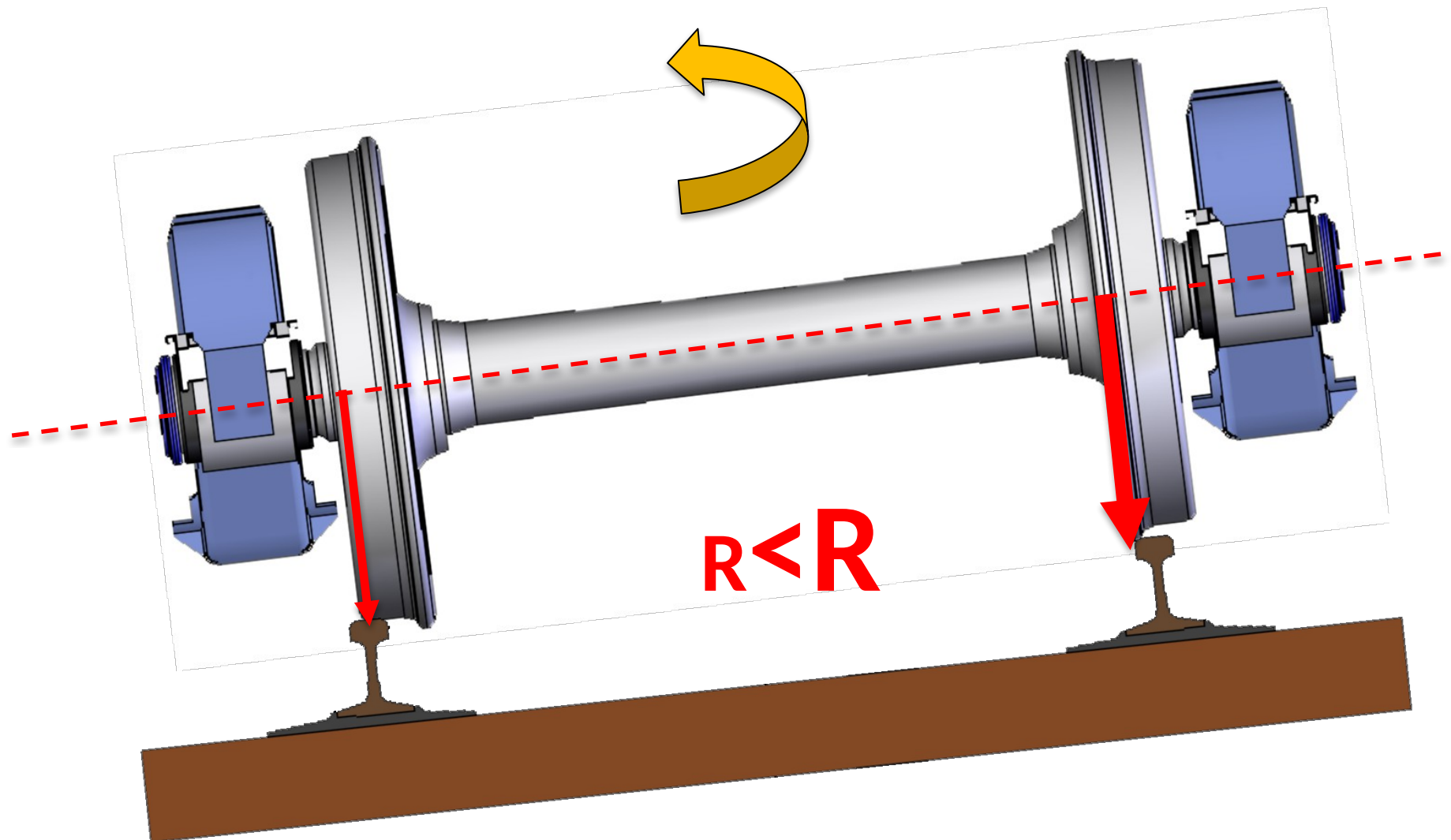
Sloped wear on
backside flange

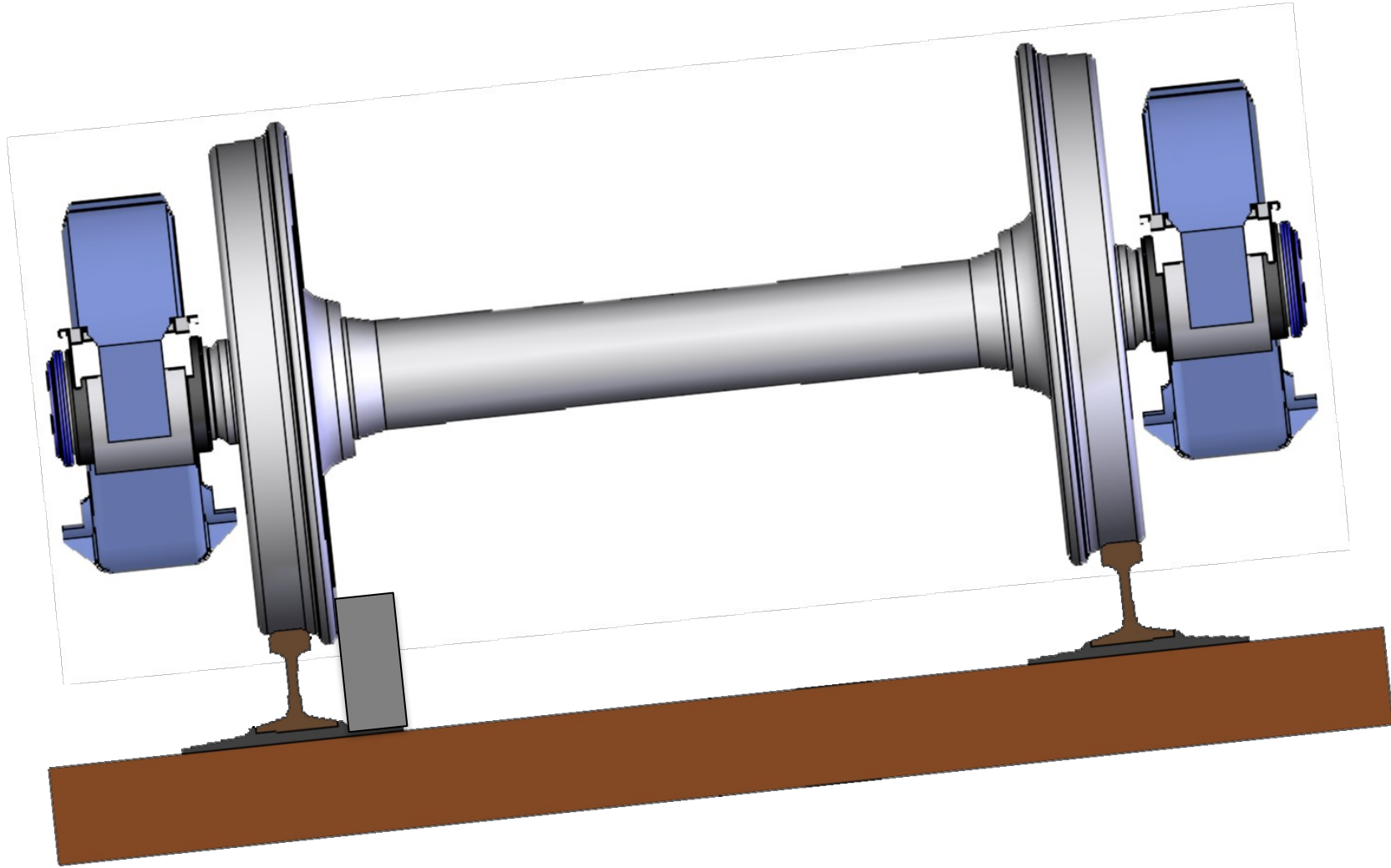




B O S T O

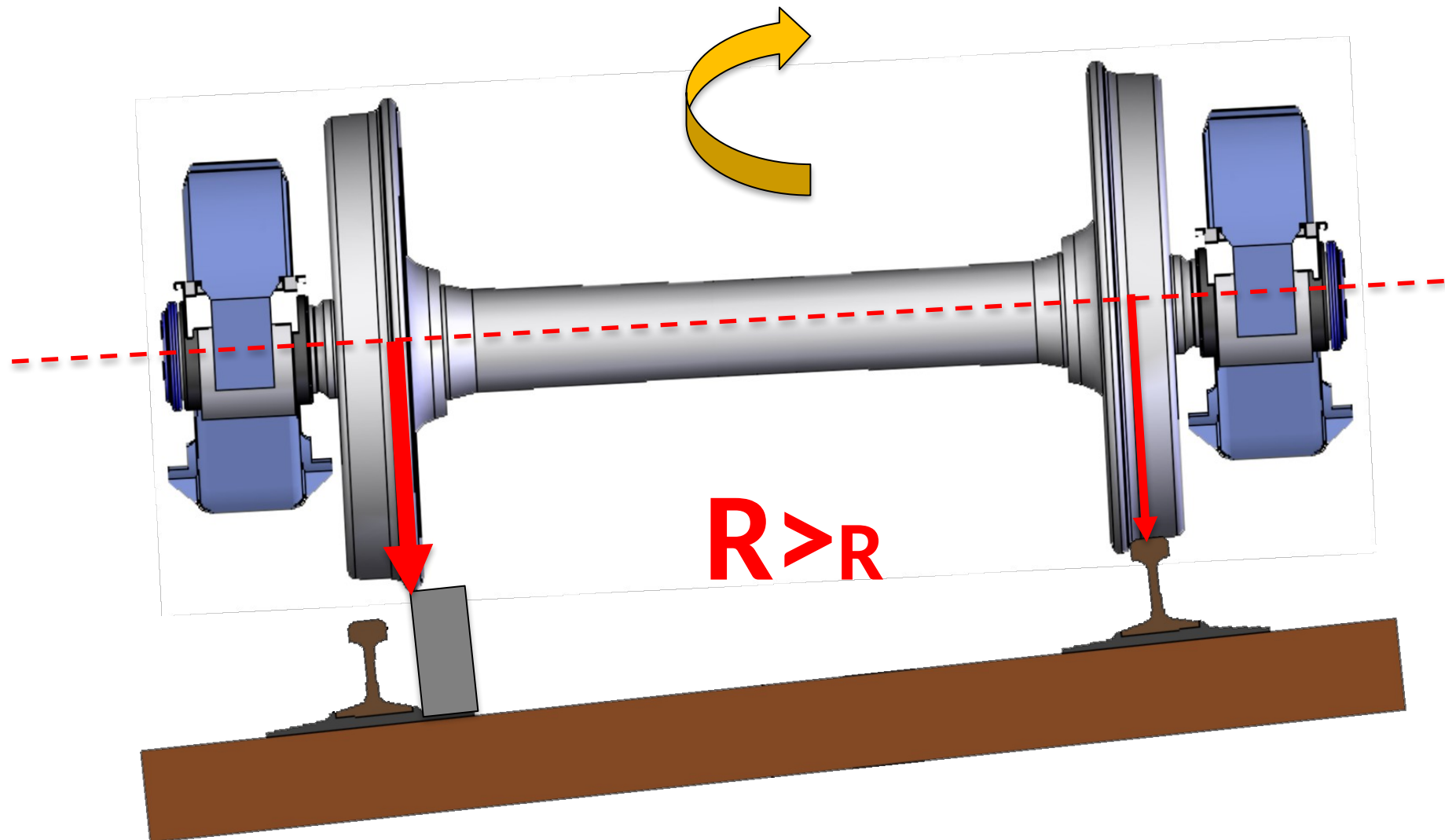






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Impact Measurement

Wheel Impact Load Detector (WILD)



Image from L.B. Foster
<https://www.youtube.com/watch?v=7CJycCggHgw>





YouTube

Search



0:23 / 0:37

FRAUSCHER



Image from Frauscher
<https://www.youtube.com/watch?v=gTfU4tGZzgo>

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3



0:24 / 0:37

FRAUSCHER



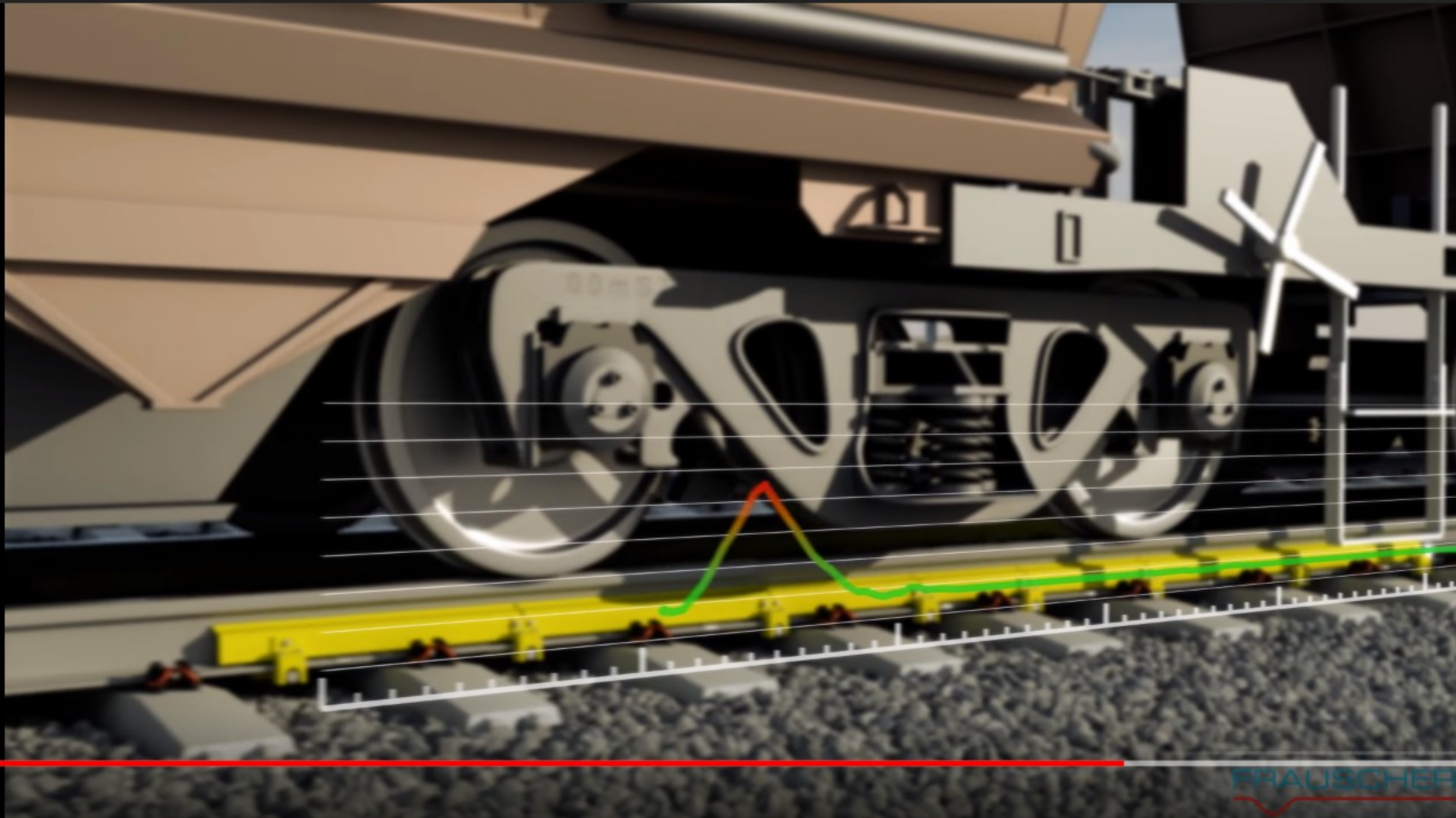
Image from Frauscher
<https://www.youtube.com/watch?v=gTfU4tGZzgo>

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0:25 / 0:37

FRAUSCHER



Image from Frauscher
<https://www.youtube.com/watch?v=gTfU4tGZzgo>

B O S T O

Example WILD Defects



Slid Flat



Shelling & Spalling



V/TI Monitor – Impact Measurement



Image from ENSCO Rail, Inc.

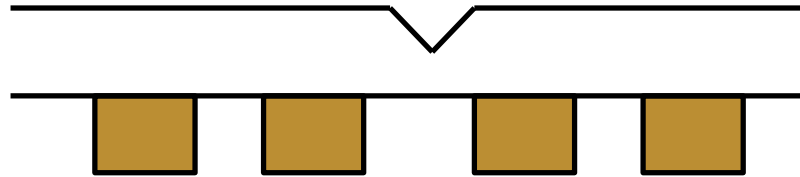


Example V/TI Monitor Axle Sensor

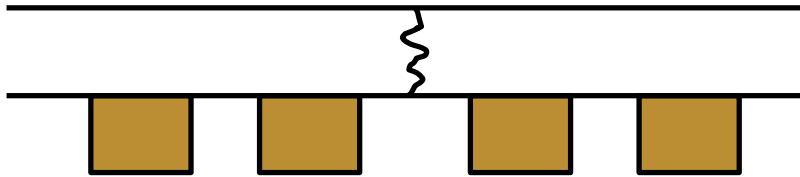


V/TI Monitor - Impact Measurement

Dent



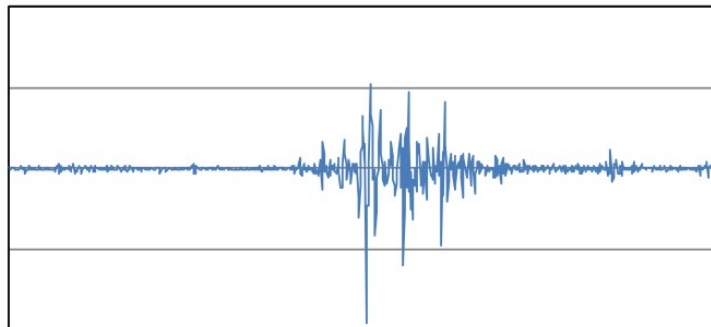
Break



Vertical acceleration
measured at axle box

Wheel/Rail Impact
acceleration measured

Peak acceleration is used to
calculate peak load.



Example V/TI Monitor Defects



Battered
Joint



Cracked/Broken
Joint Bar



Cracked/Broken
Frog



Broken Rail



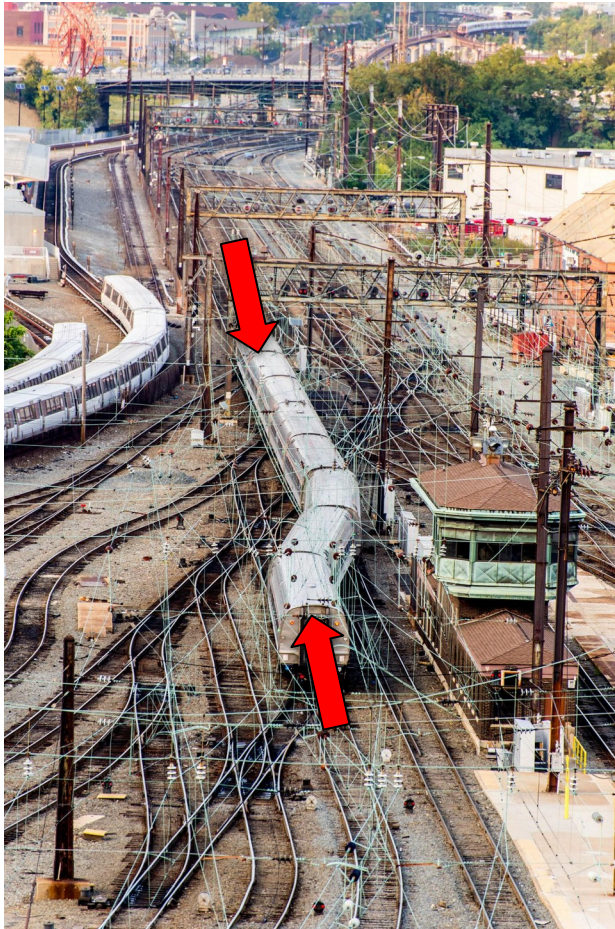
Simulations Using Measurement Data



Reasons To Do Simulations

1. New Vehicle Design and Qualification Testing
2. Aid with resolving vehicle, track, and operational issues.
3. Derailment Investigations

Simulation Types



Track-Train Dynamics Simulations
(TOES or TEDS)



Vehicle/Track Interaction Simulations
(NUCARS or VAMPIRE)

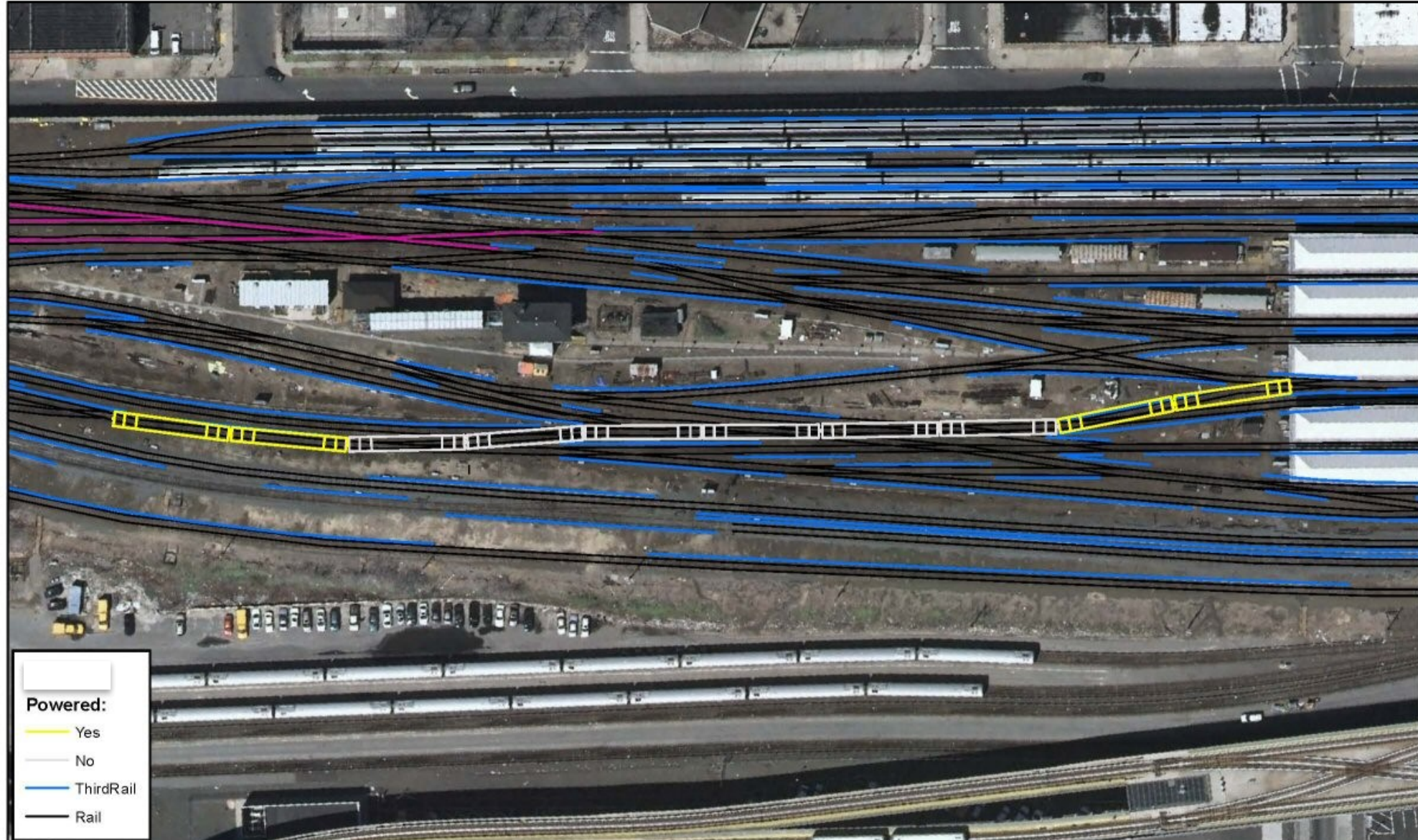


Long Gaps in Third Rail or Overhead Catenary

Causes cars to become
unpowered

Rapid throttle or
braking of powered cars
cause in-train force

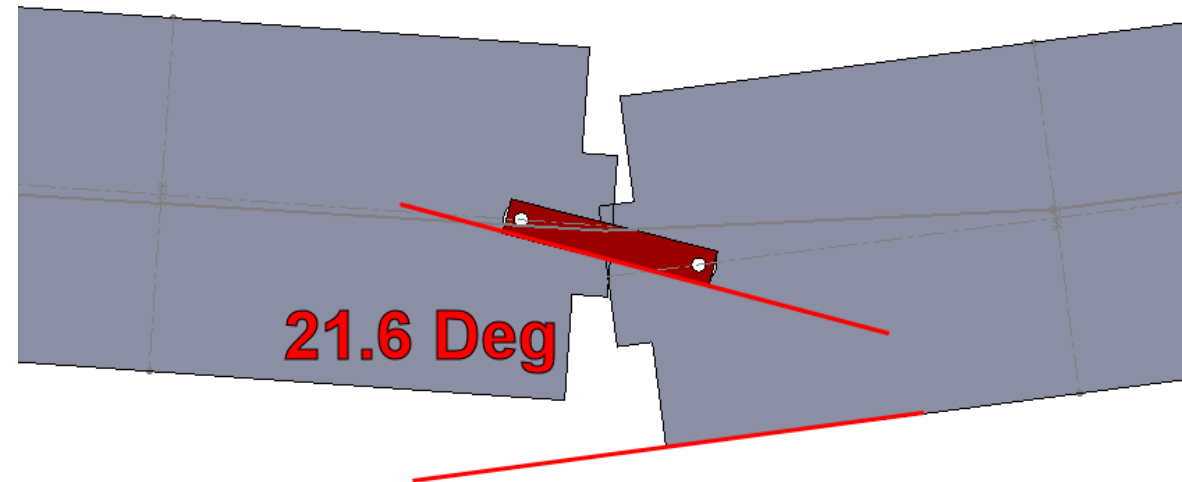
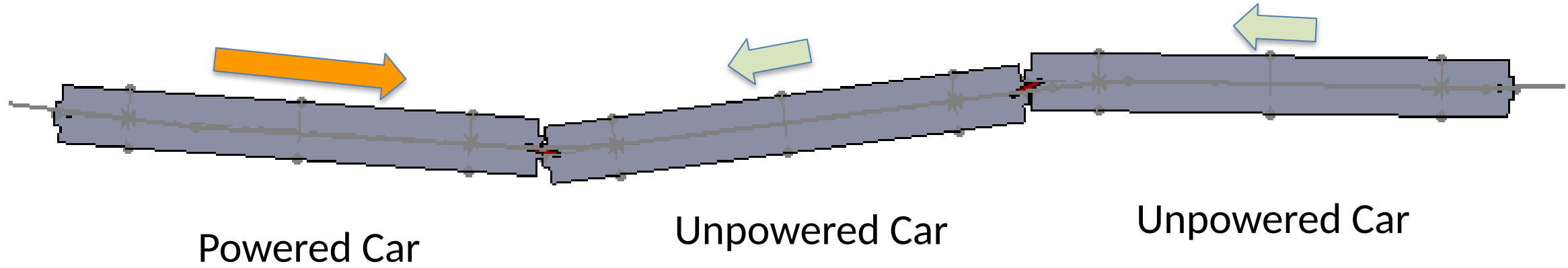




B O S T O



Train Direction



Vehicle/Track Interaction Simulations



Simulates rail vehicles travelling over track features.

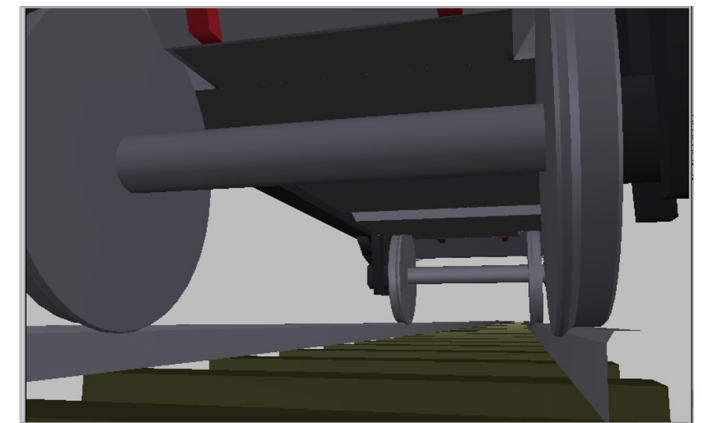
Important car inputs:

- Car type, length, weight
- Wheel profiles and diameters
- Primary and Secondary Suspension Characteristics



Important track inputs:

- Track geometry
- Rail profiles



Thank You to the Technology Vendors



Business Unit

CALIPRI



GREENWOOD ENGINEERING



QUESTIONS?

